The University of Texas at Austin
Aerospace Engineering
and Engineering Mechanics
Cockrell School of Engineering

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STD METHOD – CANTILEVER BEAM

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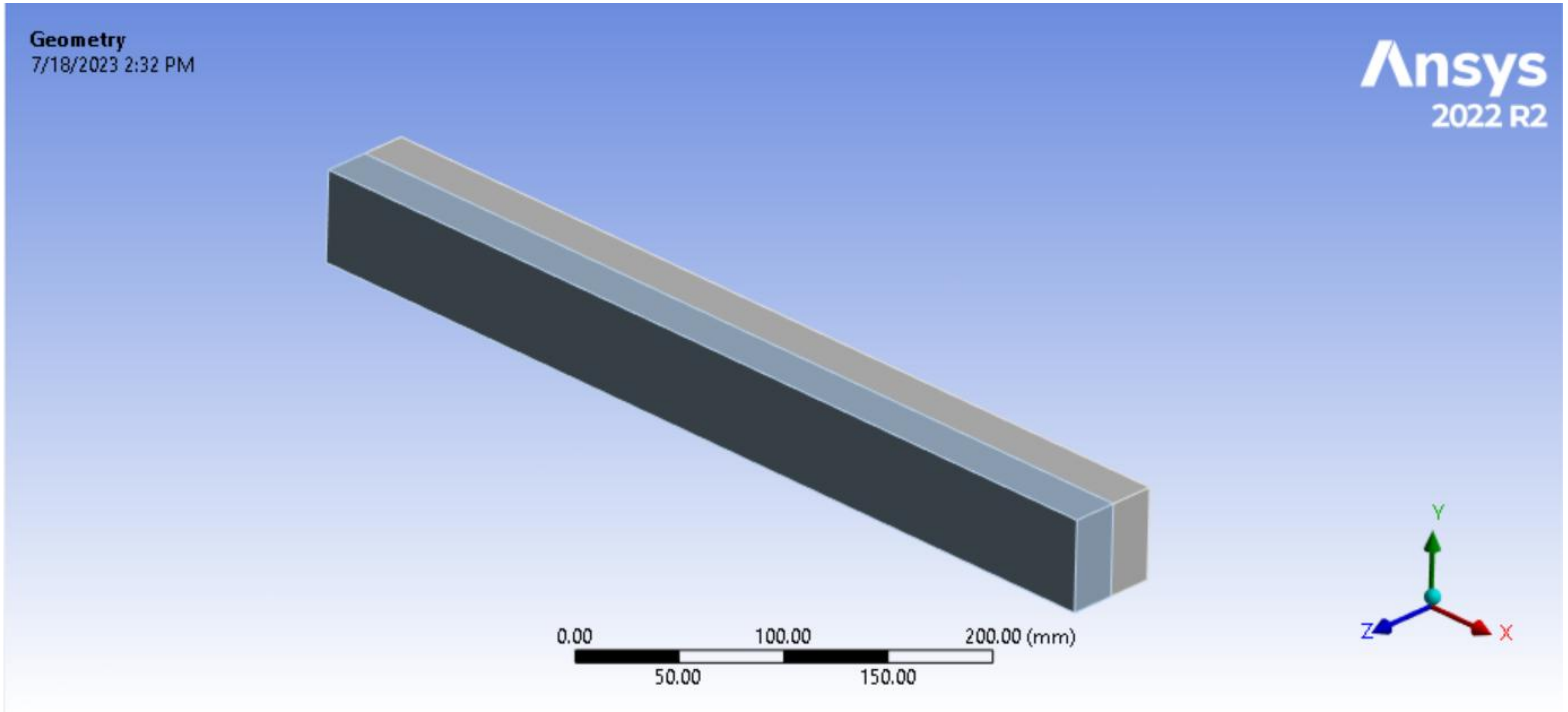
OVERVIEW:

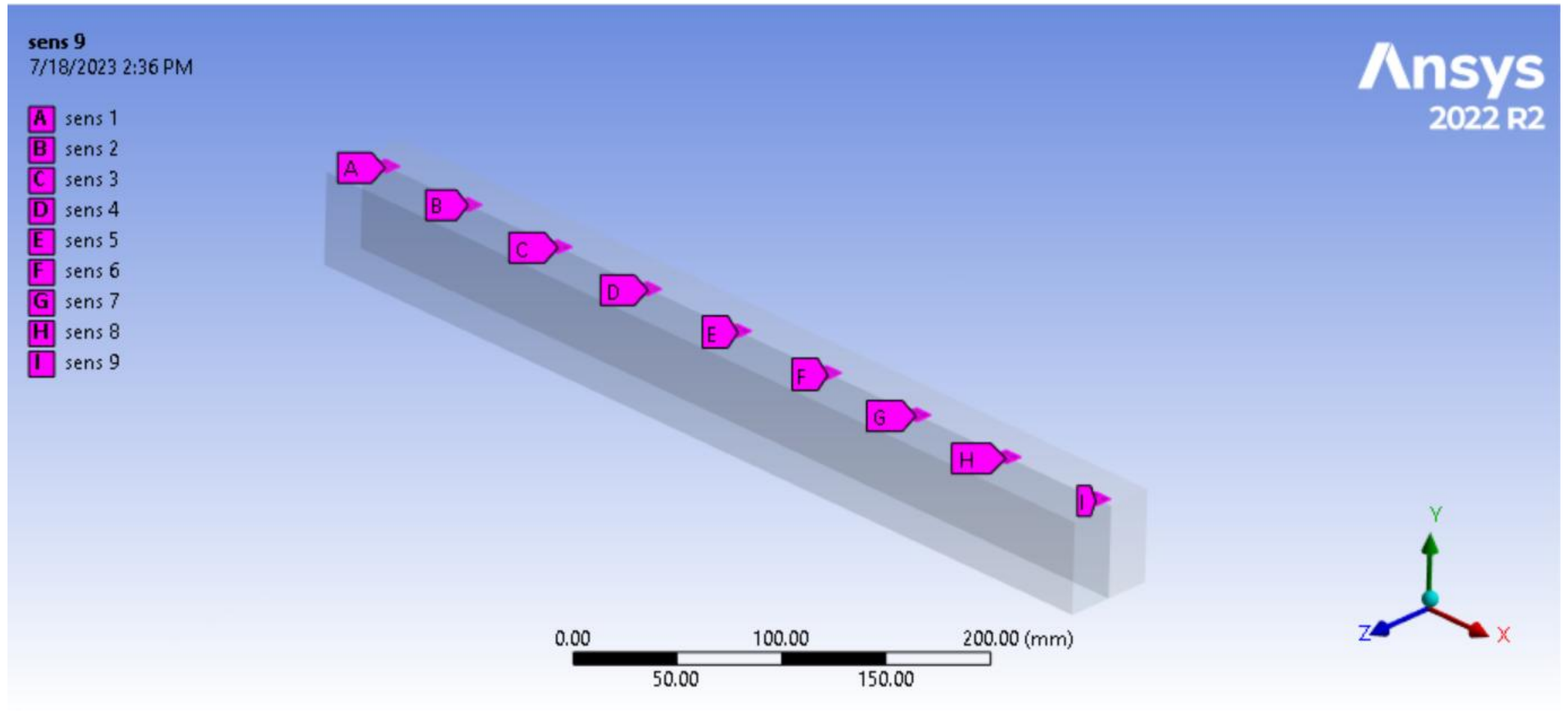
- ANSYS SIMULATED DATA COLLECTION
- DATA PROCESSING
- STD METHOD

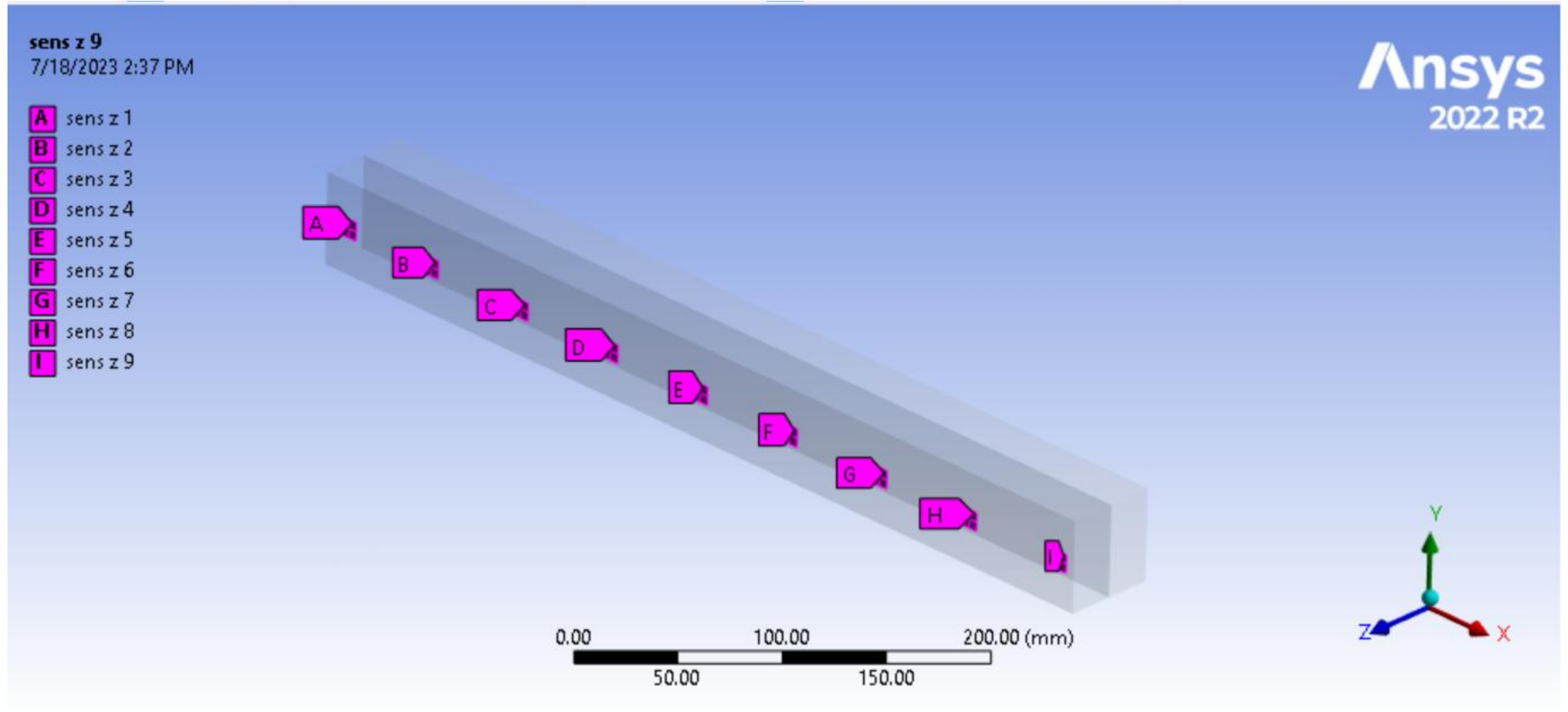
ANSYS Workbench: Setup, Meshing



- Dimensions of cantilever beam:
- 50 x 50 x 500 (in mm)
- Maximum mesh size of 5 mm
- Splitting line unimportant (done to learn SpaceClaim)
- Origin is center of leftmost square face

Virtual Sensors, Choosing Modes





- Virtual sensors created on two faces of cantilever beam, that parallel with XZ plane and XY plane
- Sensors collected data on normal strain and directional deformation in each of the X, Y and Z degrees of freedom.
- Modes chosen using participation factor for each respective DOF
- Generated modes until ratio of effective mass to total mass greater than 0.85
- For respective static structural analyses, the square face closest to sensor 1 is fixed, while the opposite face experiences a force of 10 N strictly in the X, Y, or Z direction

Data Collection

Directional Deformation Mode Shapes Y							
	Mode 2	Mode 4	Mode 8	Mode 11	Mode 13	Mode 17	Mode 20
sensor 1	0.106739	-0.74534	2.128704	-4.04579	6.261645	-8.54397	10.71414
sensor 2	1.151002	-6.24259	14.28207	-21.0883	24.086	-22.0843	15.55544
sensor 3	3.606081	-15.1784	24.60692	-19.4313	2.562029	14.48448	-20.1312
sensor 4	7.166436	-22.2472	19.03024	7.070196	-21.3038	6.325975	16.58727
sensor 5	11.56119	-23.7315	0.008345	22.08816	2.577489	-20.7874	-3.80844
sensor 6	16.53761	-17.981	-17.606	4.681525	21.30822	12.31809	-10.6942
sensor 7	21.87234	-5.55769	-19.0142	-18.4897	-5.7387	10.45181	20.27588
sensor 8	27.38417	11.29238	-1.04478	-10.8267	-16.782	-18.6365	-16.4367
sensor 9	32.94768	29.78904	26.84178	23.93575	21.29759	18.88363	16.62317
Strain Mode Shapes							
	Mode 2	Mode 4	Mode 8	Mode 11	Mode 13	Mode 17	Mode 20
sensor 1	-0.01083	0.056228	-0.12349	0.174892	-0.19137	0.164607	-0.09604
sensor 2	-0.00963	0.022778	0.014151	-0.12924	0.273299	-0.36064	0.313859
sensor 3	-0.00768	-0.01259	0.107057	-0.16642	0.013988	0.318425	-0.5465
sensor 4	-0.0058	-0.03826	0.090836	0.078038	-0.32686	0.13447	0.428141
sensor 5	-0.00405	-0.0497	-0.00986	0.219108	0.034883	-0.42546	-0.10175
sensor 6	-0.0025	-0.04594	-0.11044	0.038514	0.312194	0.246463	-0.28503
sensor 7	-0.00125	-0.03064	-0.13241	-0.20983	-0.11144	0.189666	0.504788
sensor 8	-0.00038	-0.01173	-0.06927	-0.18747	-0.34295	-0.48115	-0.53814
sensor 9	-4.20E-06	-0.00025	-0.00219	-0.00858	-0.02253	-0.04705	-0.08497

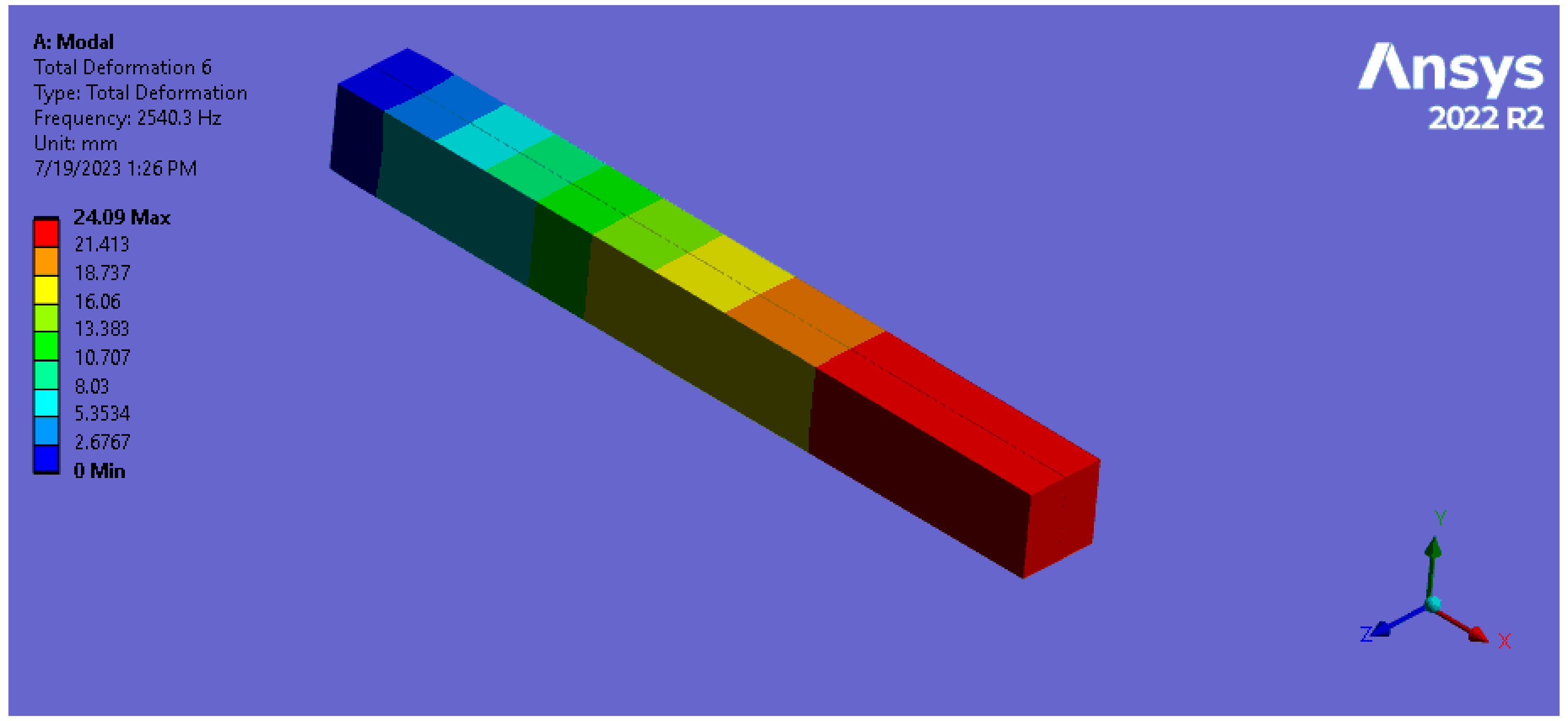
Static Deformation
7.87E-07
3.84E-06
7.22E-06
1.06E-05
1.40E-05
1.74E-05
2.07E-05
2.41E-05
2.75E-05
Static Strain
-3.11E-06
-2.91E-06
-2.50E-06
-2.10E-06
-1.69E-06
-1.28E-06
-8.79E-07
-4.73E-07
-9.72E-08

(Left) Y-deformation mode shapes of modes with high participation factors in the Y-direction, as well as strain mode shape for the same modes
(Right) An example of collected data for static deformation in one direction, and static strain (in this case, X deformation from an X-force, and normal strain from the same force)

- In modal analysis, collected average directional deformation for all DOF for each important mode (EX: for the X DOF, since modes 6 and 15 had significant participation factors, mode shapes for said modes are generated for X, Y and Z DOFs to be used in the STD Method), also did the same for normal strain mode shapes
- In static structural analysis, collected average directional deformation and average normal strain for each DOF, at each sensor, for all forces in each of the X, Y, or Z directions
- Tabulated and organized by sensor, DOF, and either mode or force direction, depending on the type of analysis

Modes Chosen: X-DOF

6



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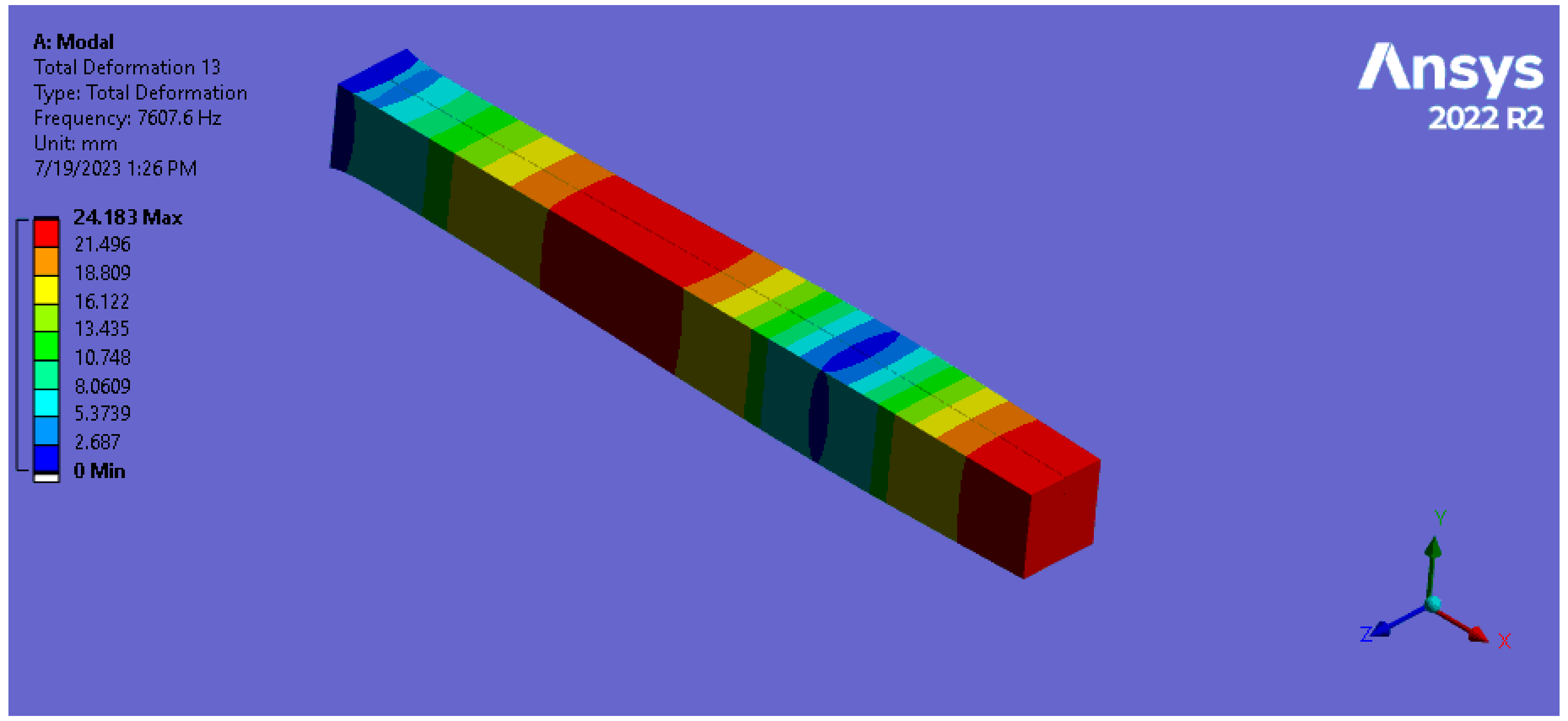

Modes Chosen: Y-DOF

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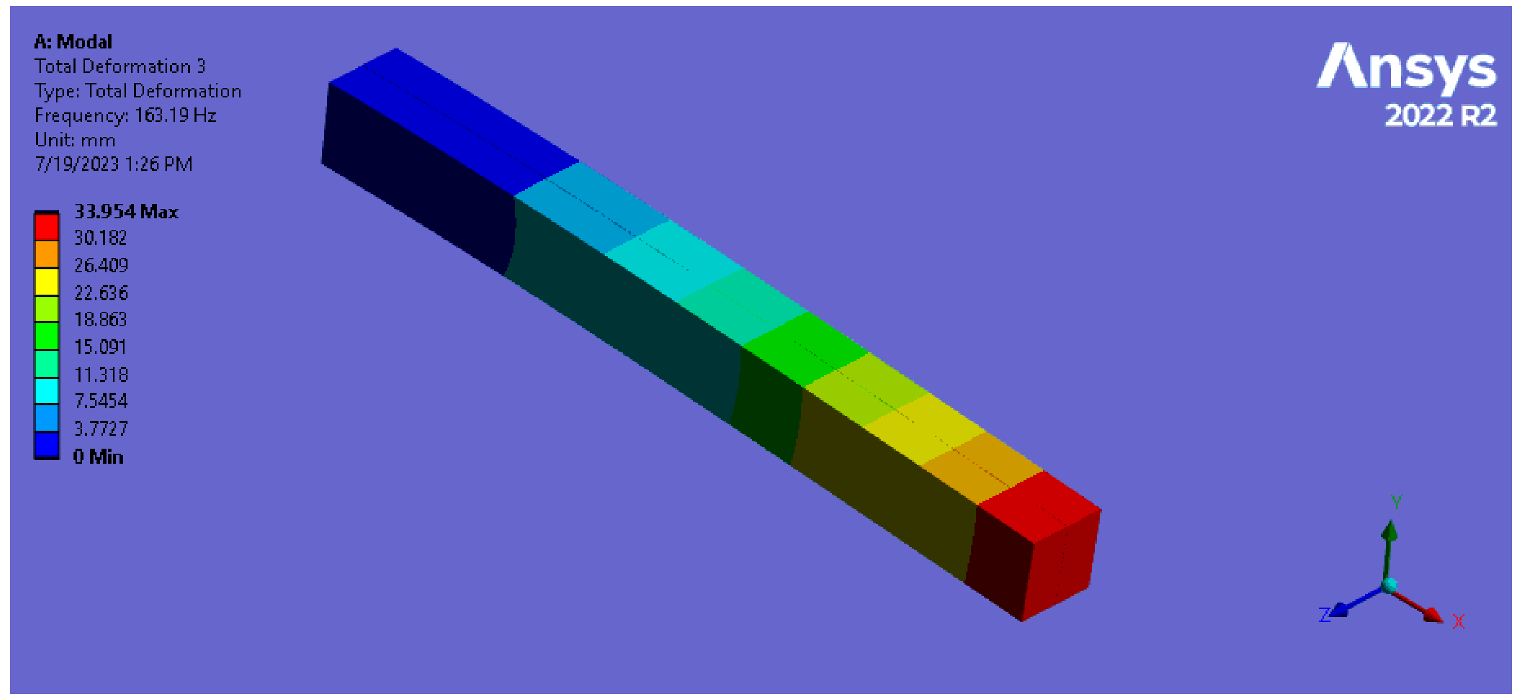

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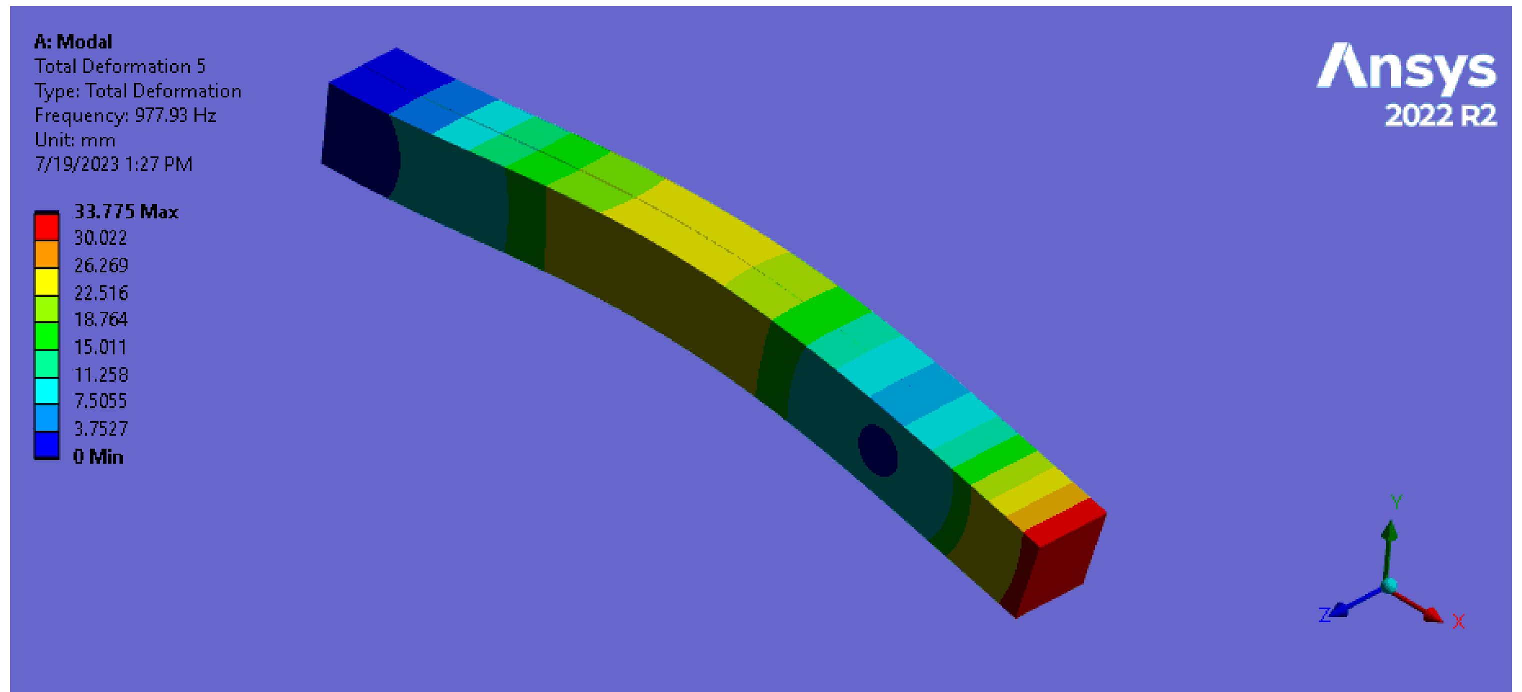

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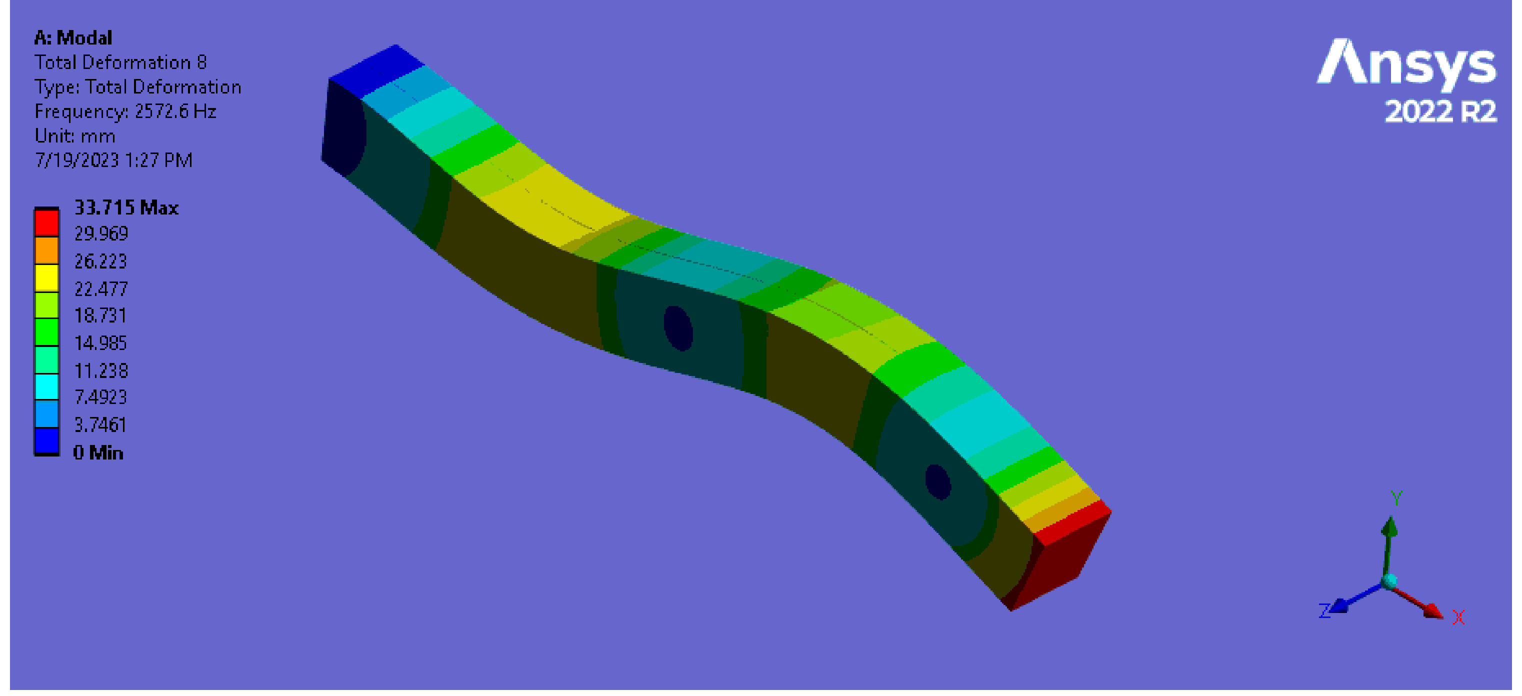

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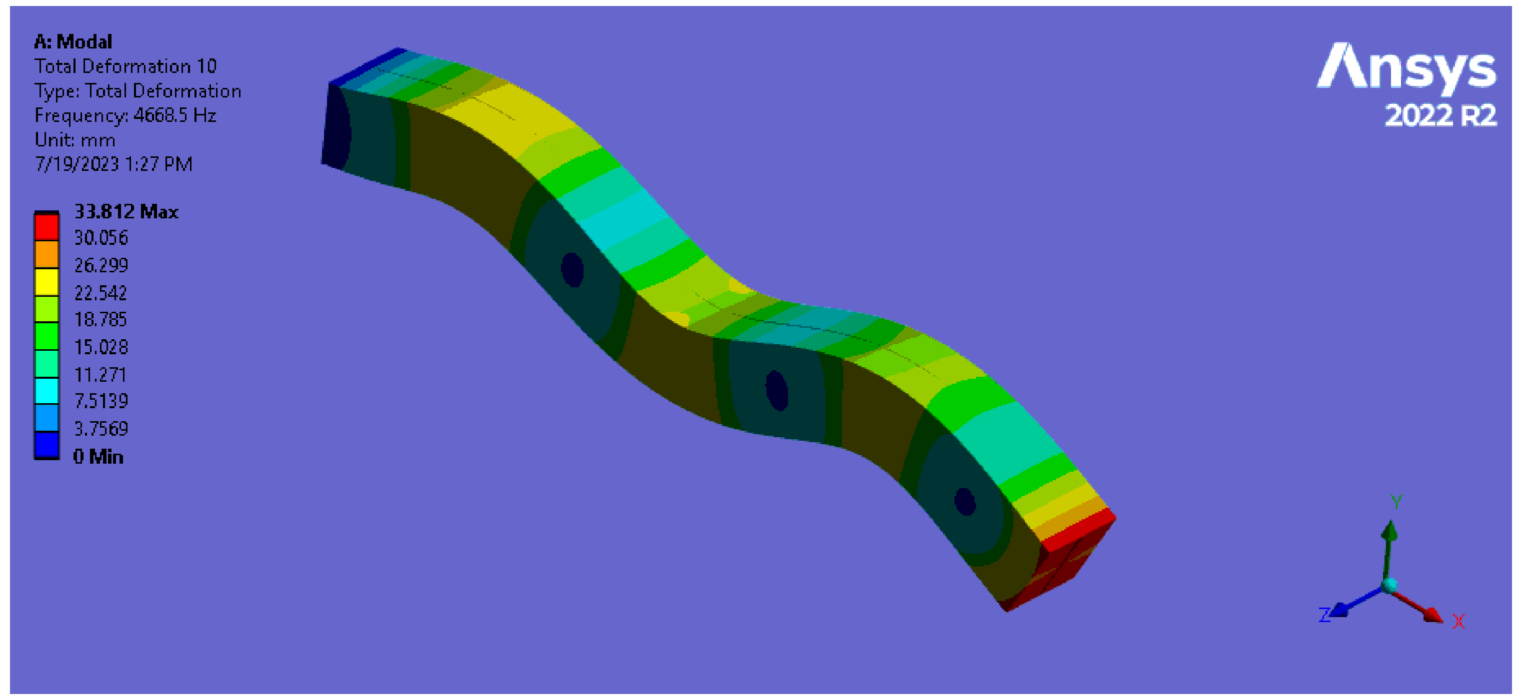

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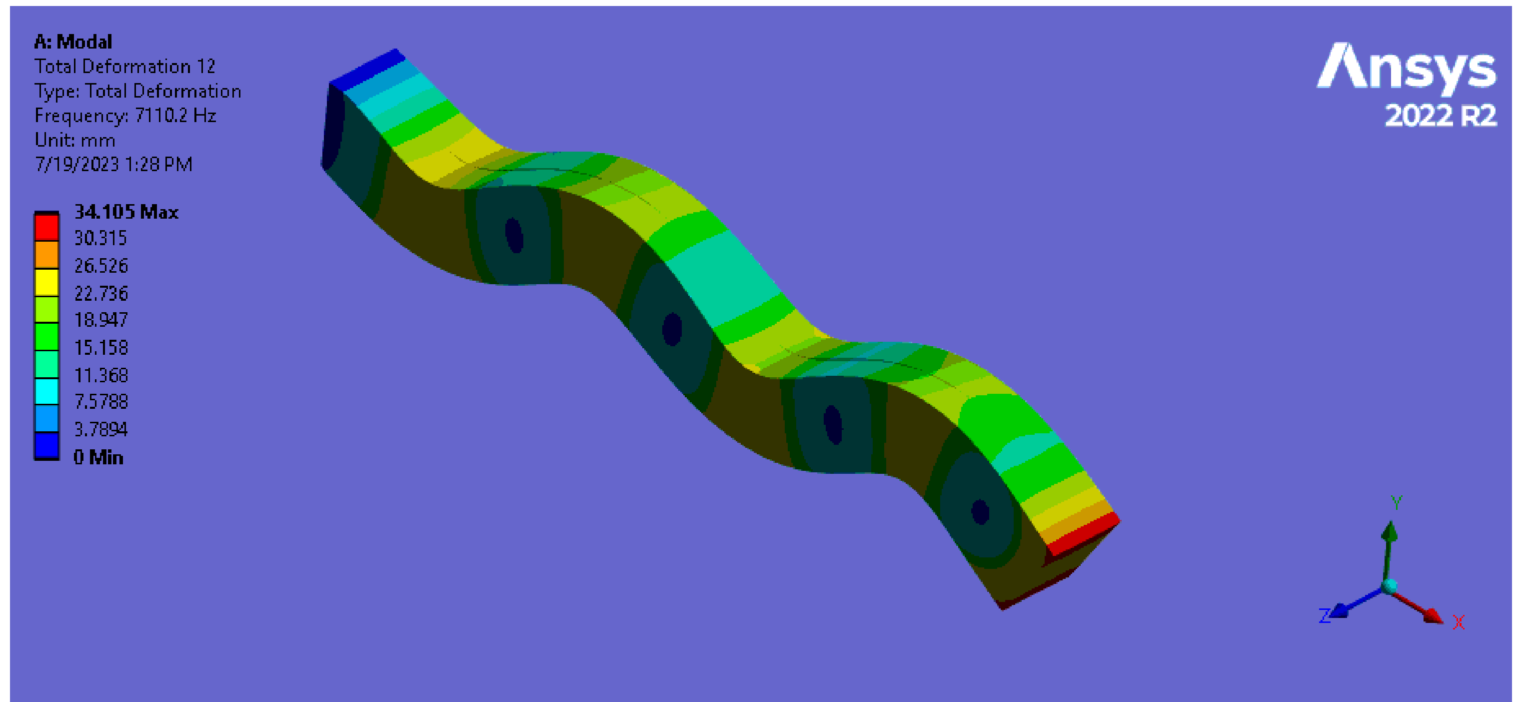

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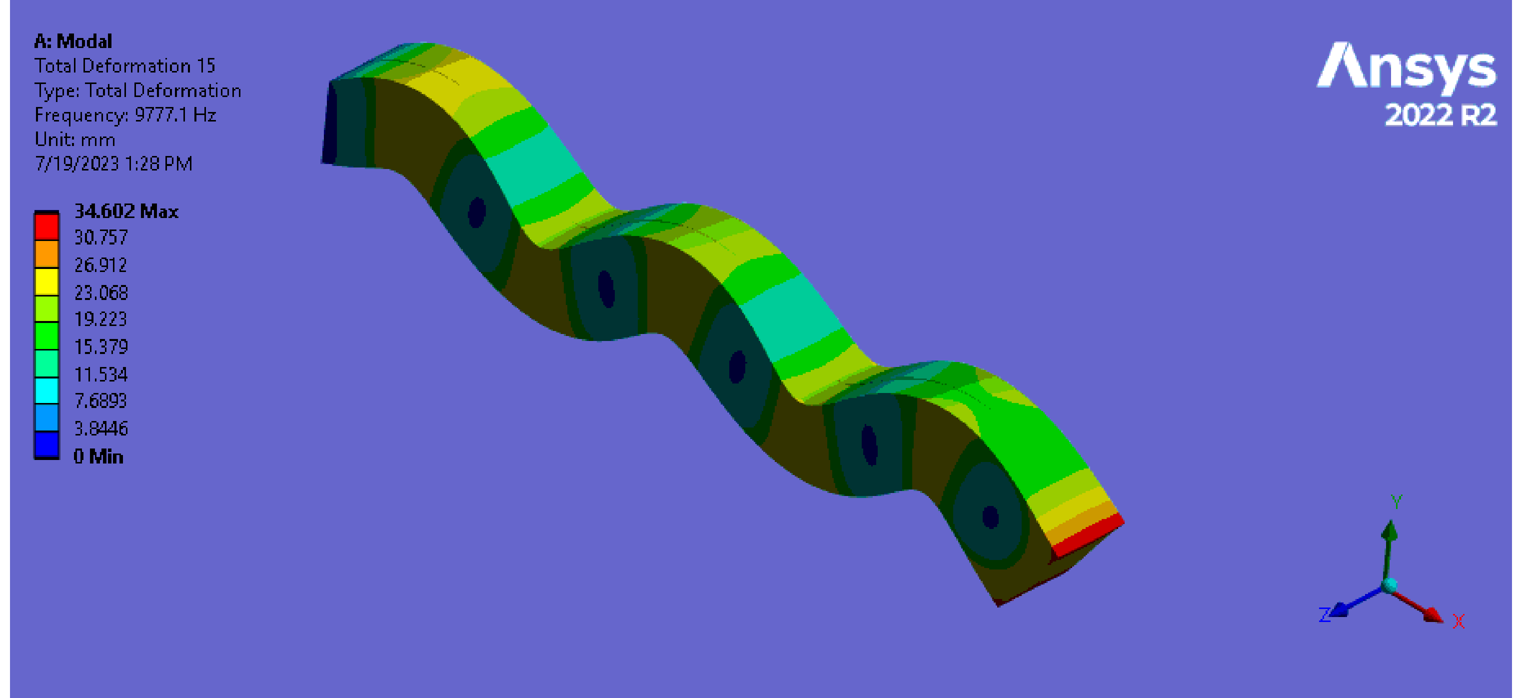

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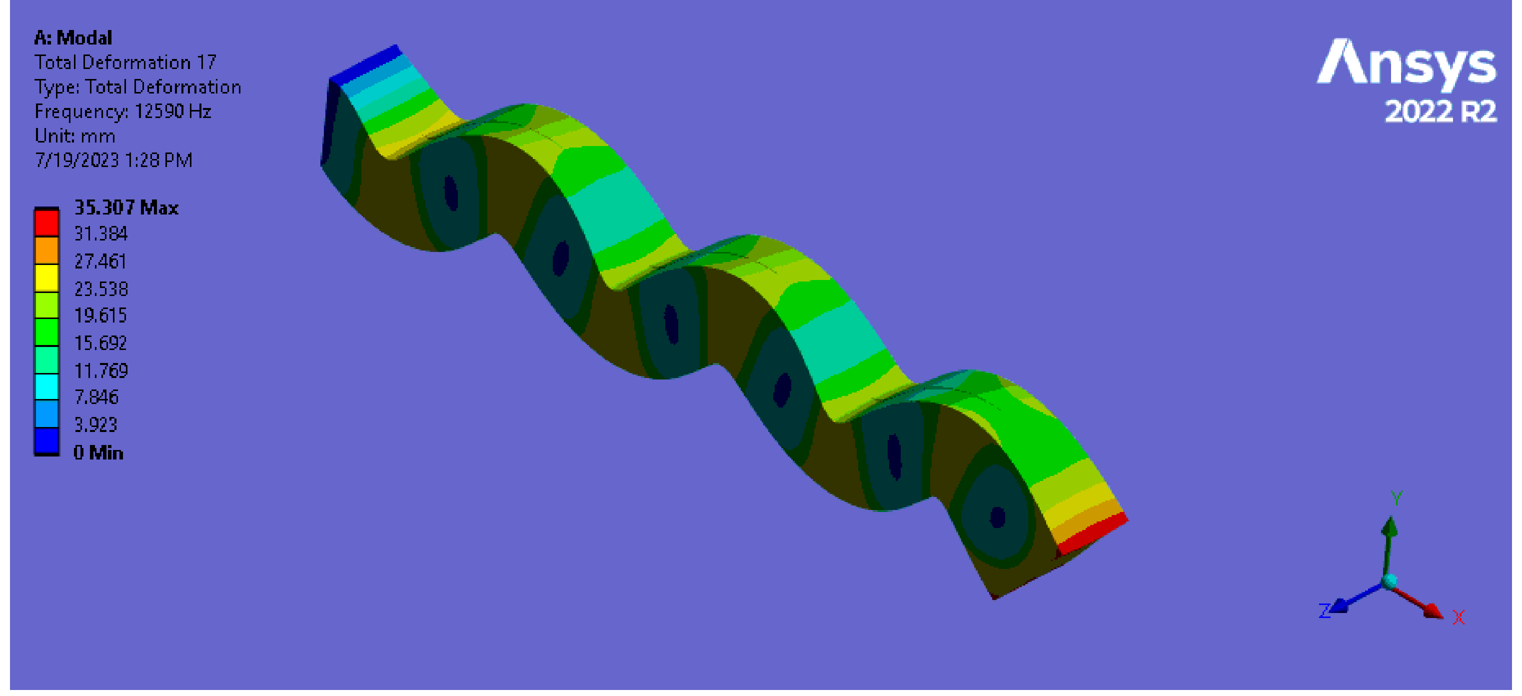
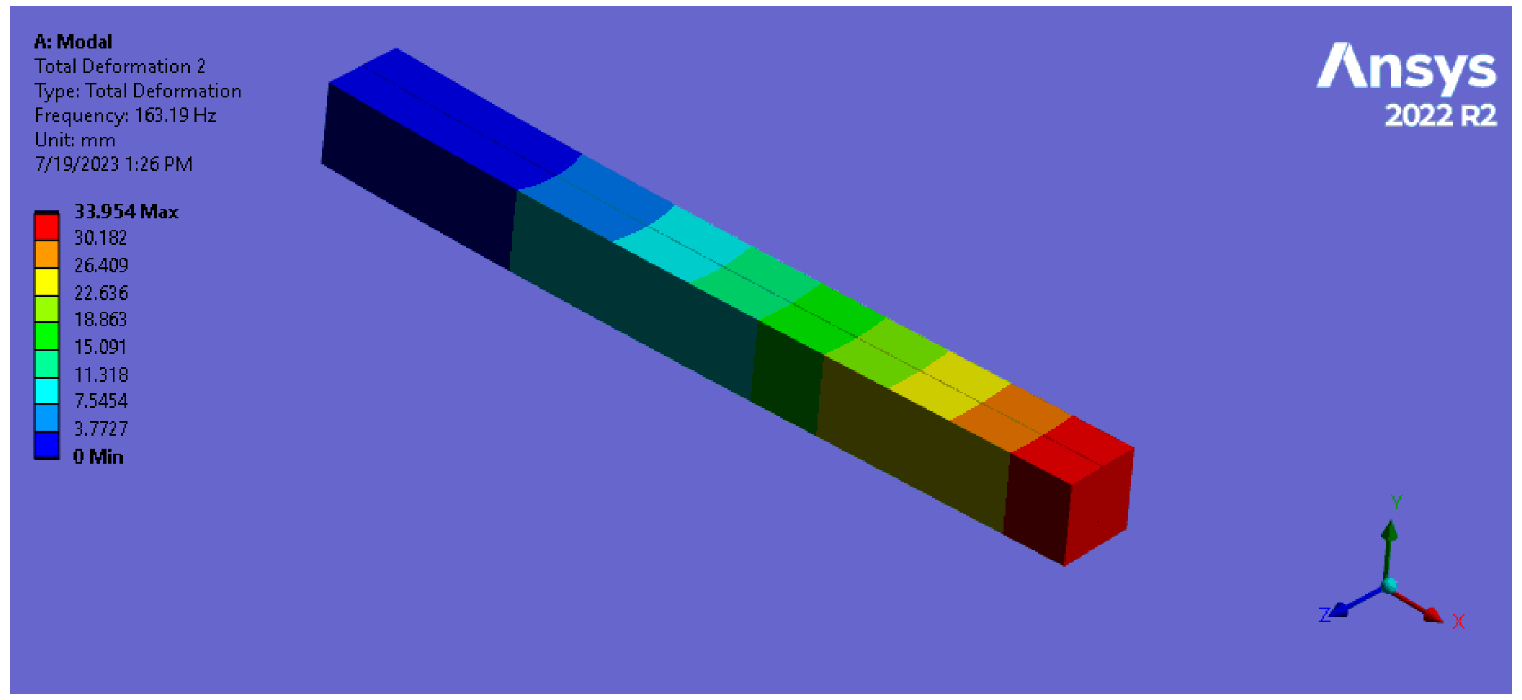


Modes Chosen: Z-DOF

1



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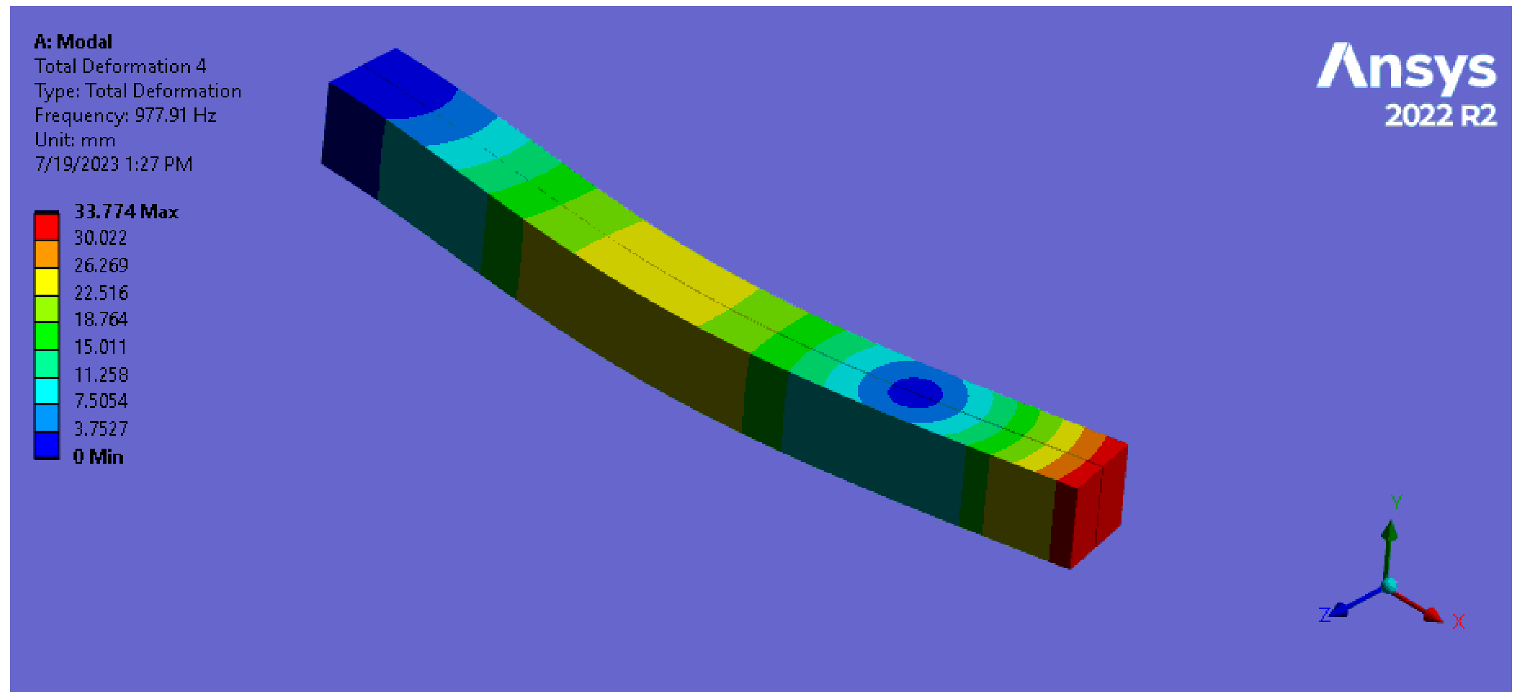

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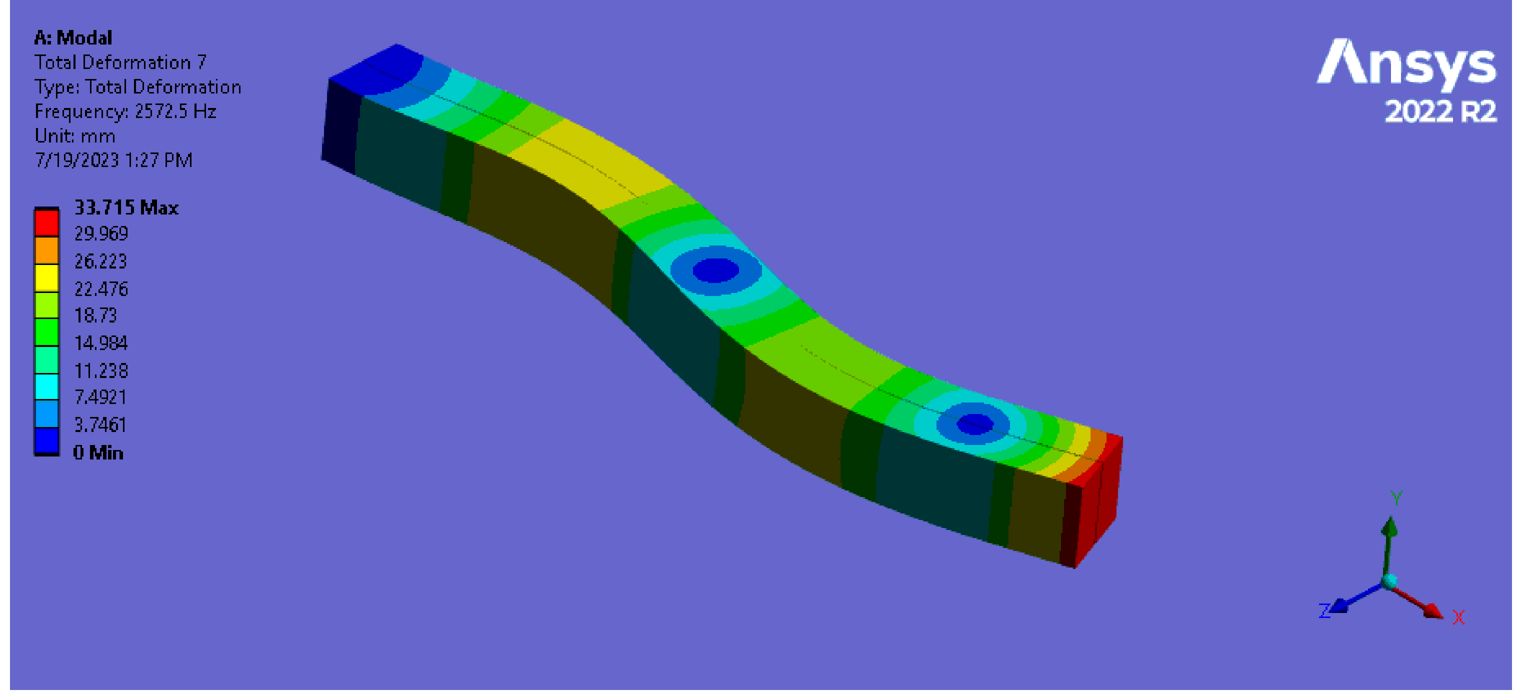

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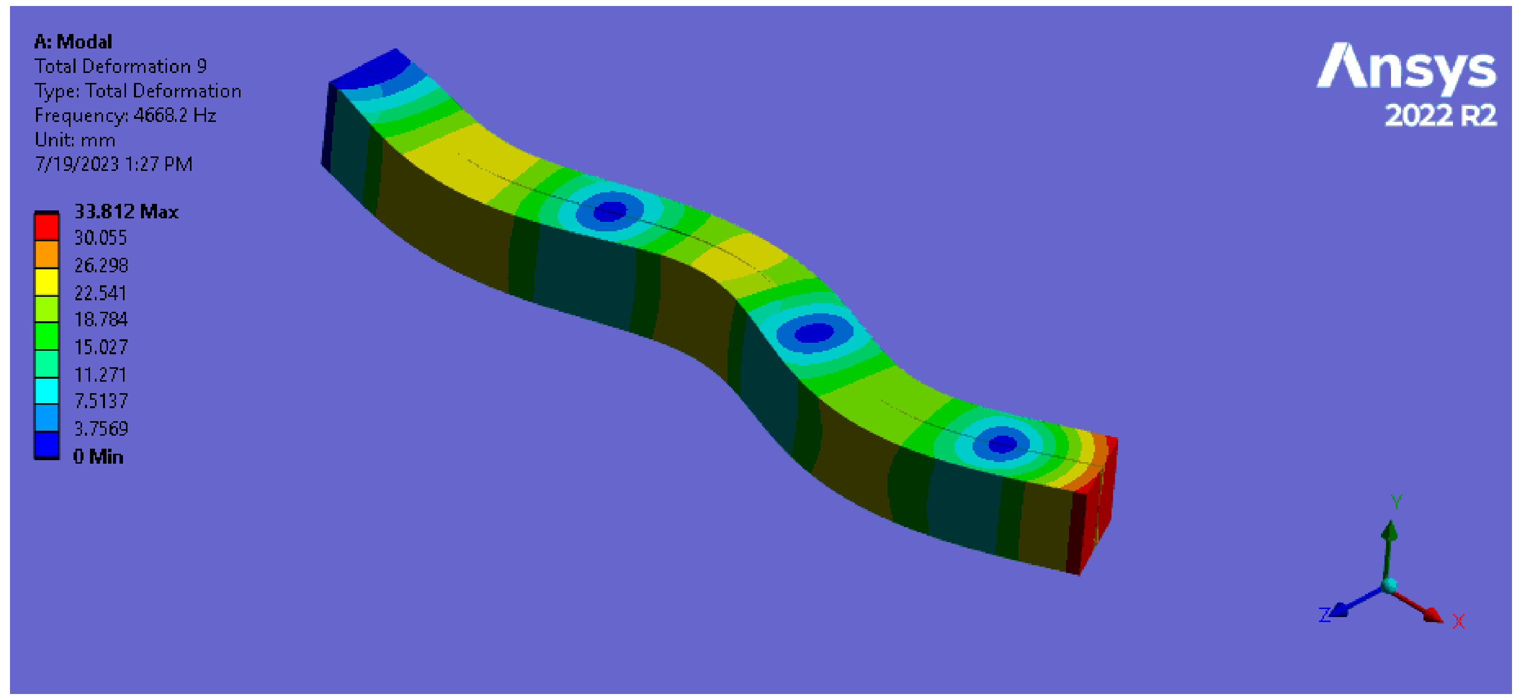

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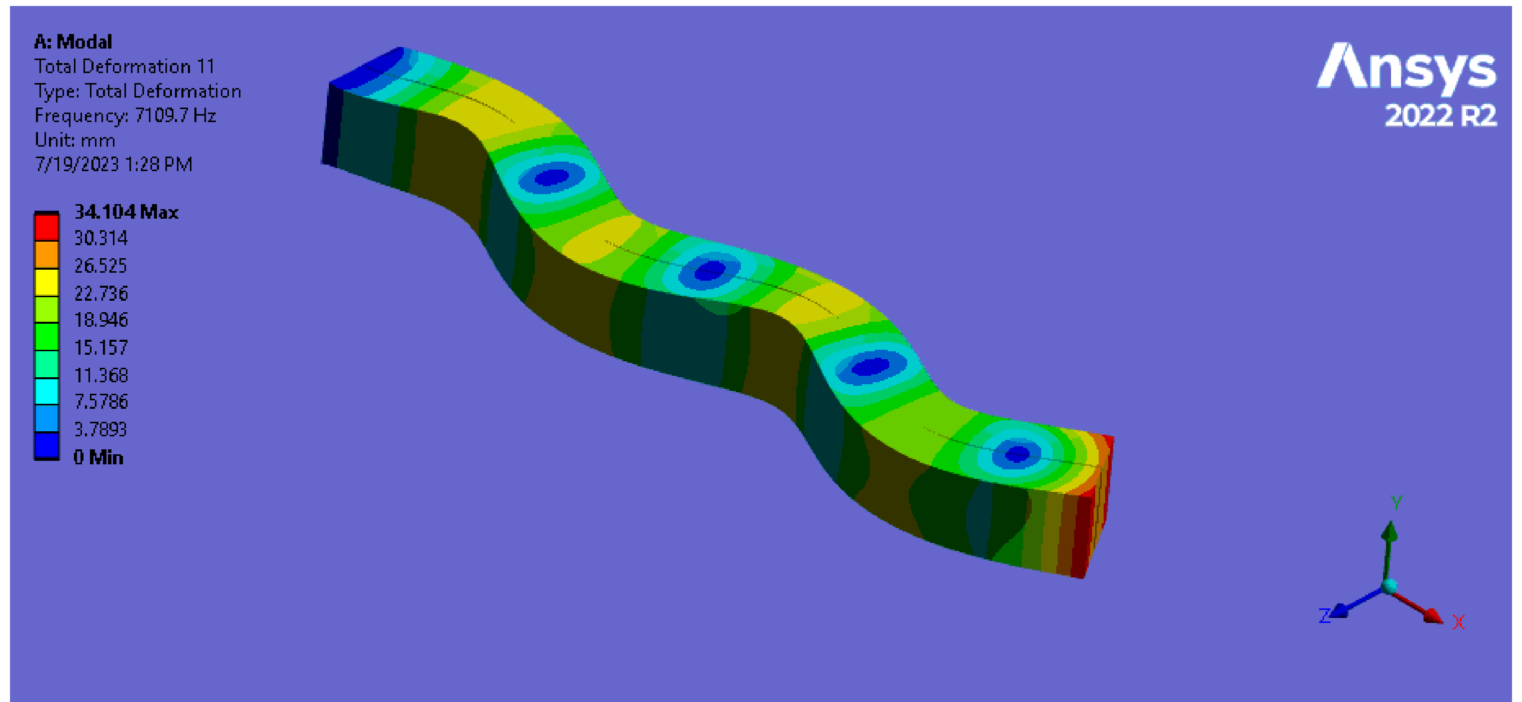

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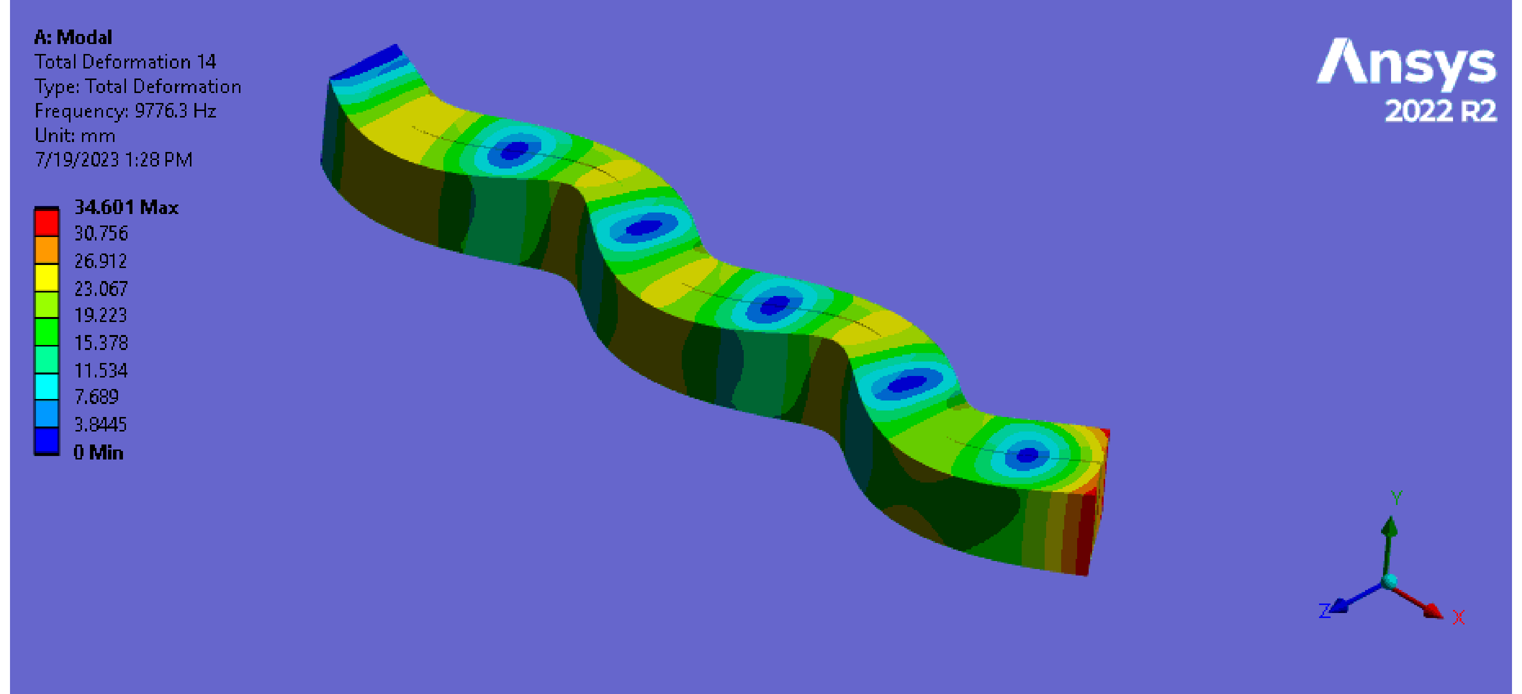

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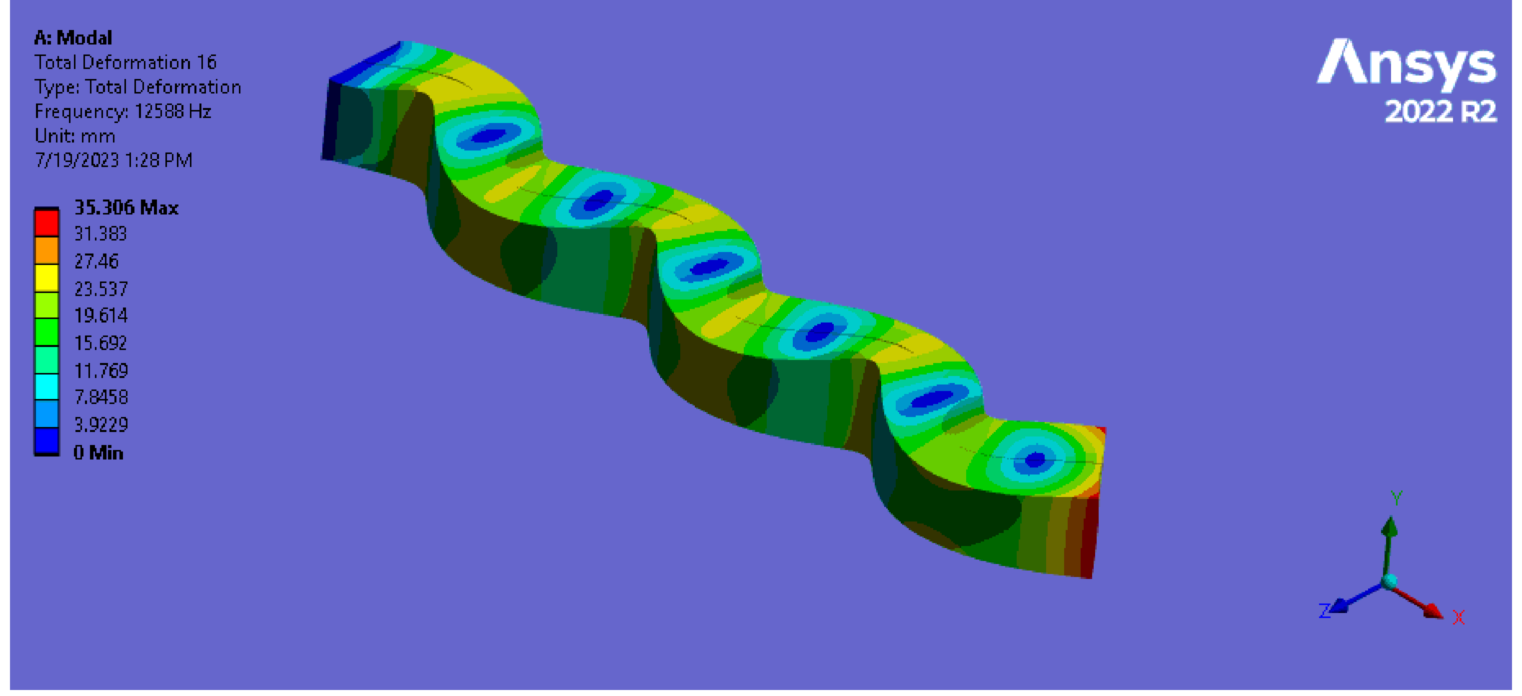


STD Method: Overview

$$\{u\} = [\Phi]\{\xi\}$$

$$\{\epsilon\} = \frac{\partial u}{\partial x} = \frac{\partial}{x}[\Phi]\{\xi\} = [\Psi]\{\xi\}$$

$$\{\xi\} = [[\Psi]^T[\Psi]]^{-1}[\Psi]^T\{\epsilon\}$$

$$\{u\} = [\Phi][[\Psi]^T[\Psi]]^{-1}[\Psi]^T\{\epsilon\}$$

- From “Fiber-Optics-Based Aeroelastic Shape Sensing” by Freydin et al.
- Going from relation between deformation and deformation mode shapes, as well as strain and strain mode shapes, and ending with an equation using said mode shapes to find deformation from strain

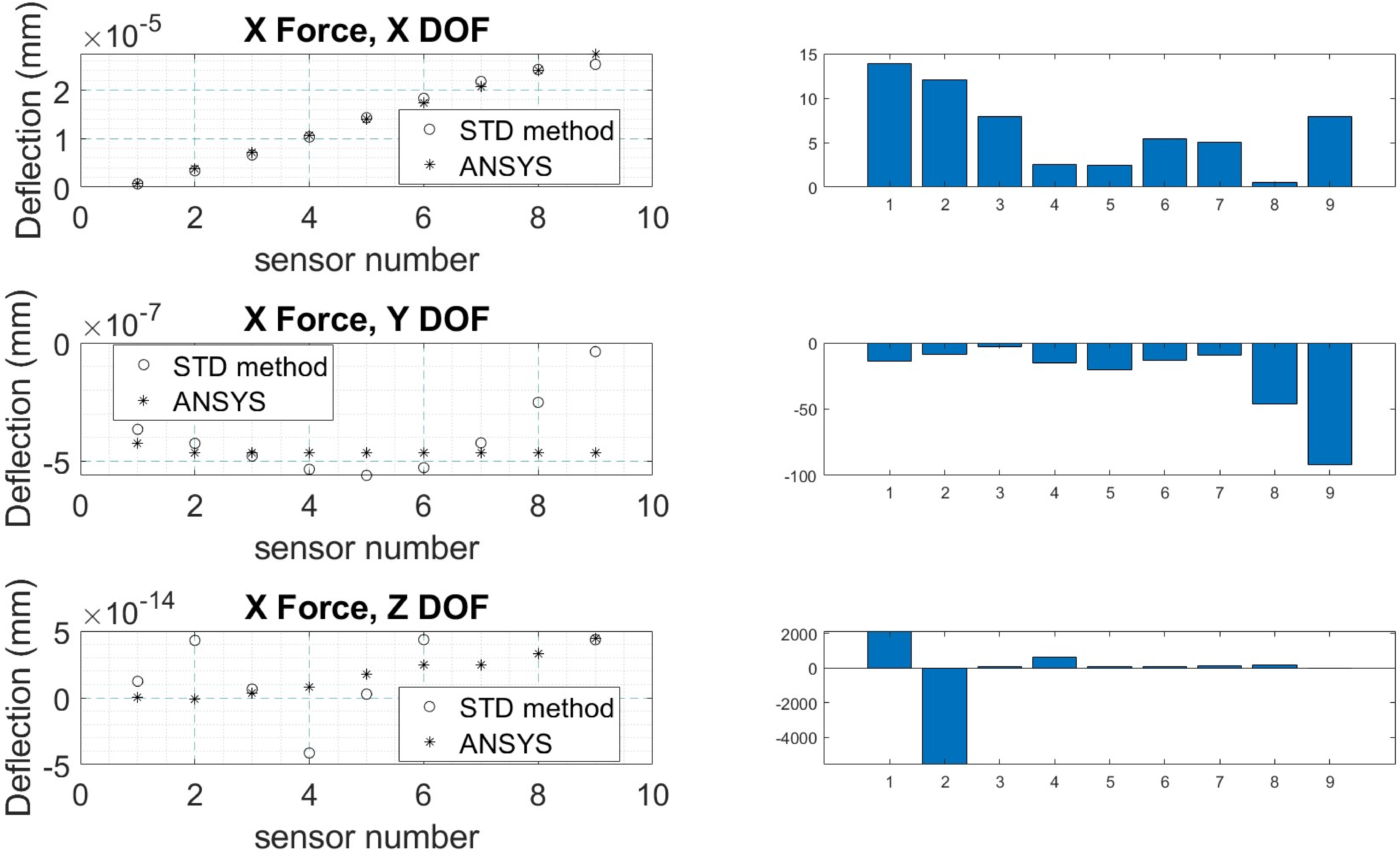
Setting up Matrices

MODE A-W (or with Static, only one column utilized)		
X-deformation for sensor 1, mode A	…	X-deformation for sensor 1, mode W
Y-deformation for sensor 1, mode A	…	Y-deformation for sensor 1, mode W
Z-deformation for sensor 1, mode A	…	Z-deformation for sensor 1, mode W
…	…	…
X-deformation for sensor n, mode A	…	X-deformation for sensor n, mode W
Y-deformation for sensor n, mode A	…	Y-deformation for sensor n, mode W
Z-deformation for sensor n, mode A	…	Z-deformation for sensor n, mode W

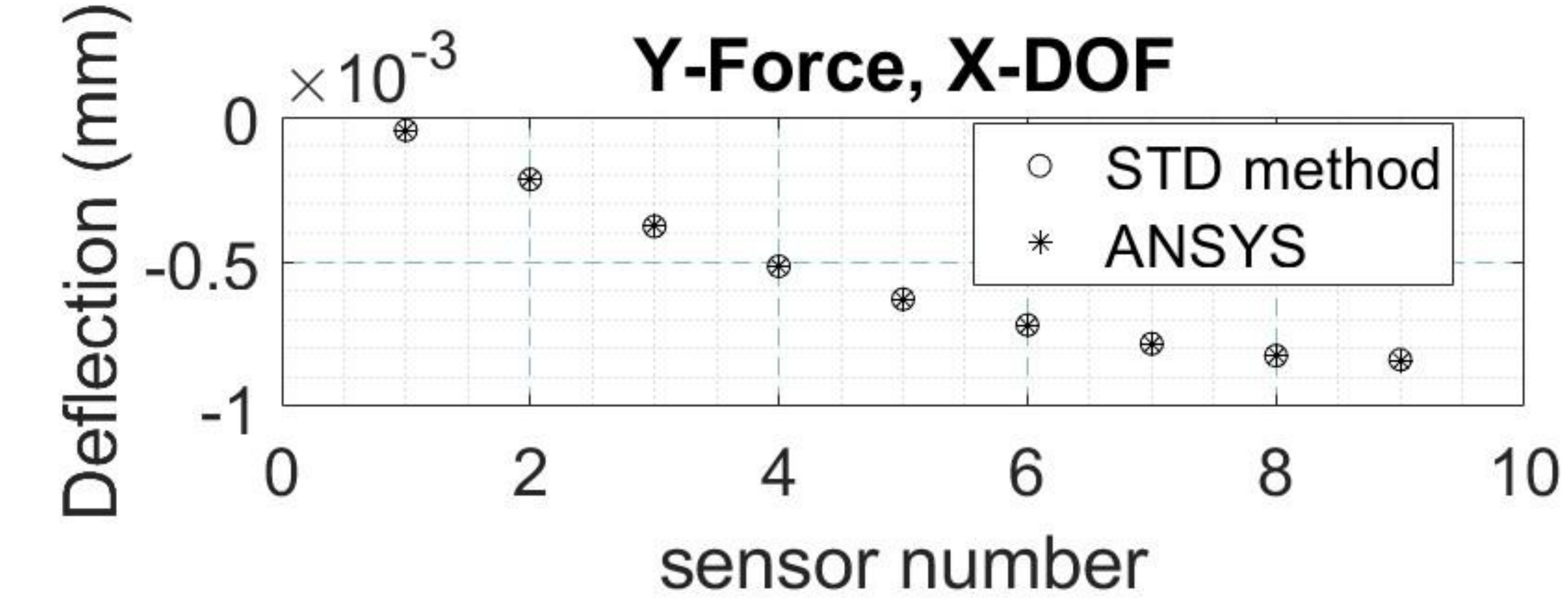
(Figure) Simplified representation of how modal and static matrices formatted for data processing

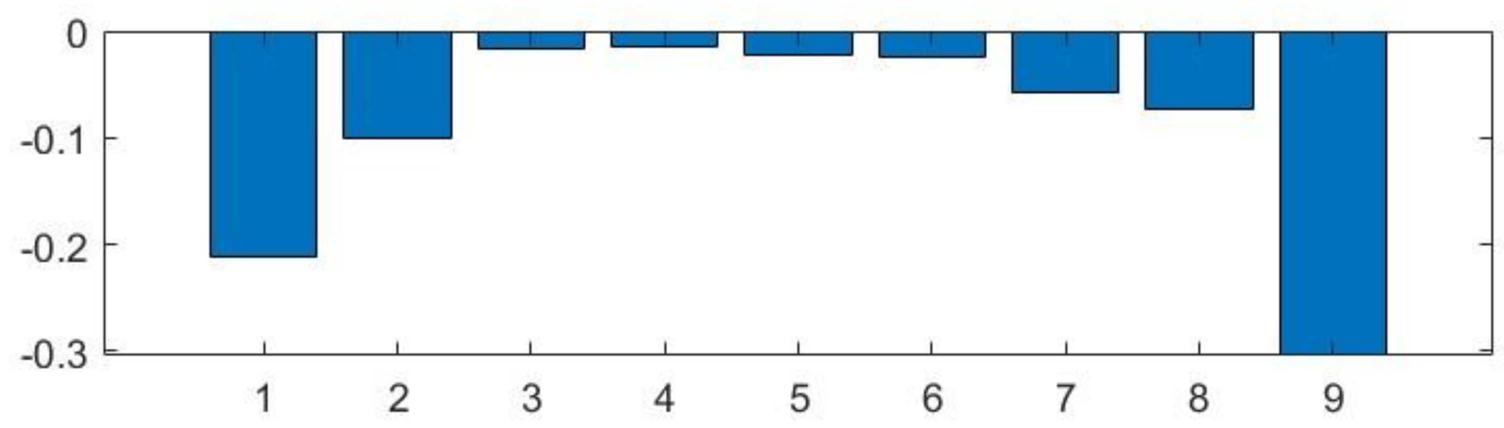
- Deformation mode shape matrices created by having each 3 rows representing a sensor
- Each row is X, Y, or Z for said sensor
- Each column represents one mode
- Strain mode shape matrices created similarly except the Y and Z strain is zero as only normal strain in the X-direction is measured
- Static deformation is a single column vector arranged in the same way (arrangements of 3 rows for each node)
- Static strain similar to static deformation, except again, without Y or Z strains

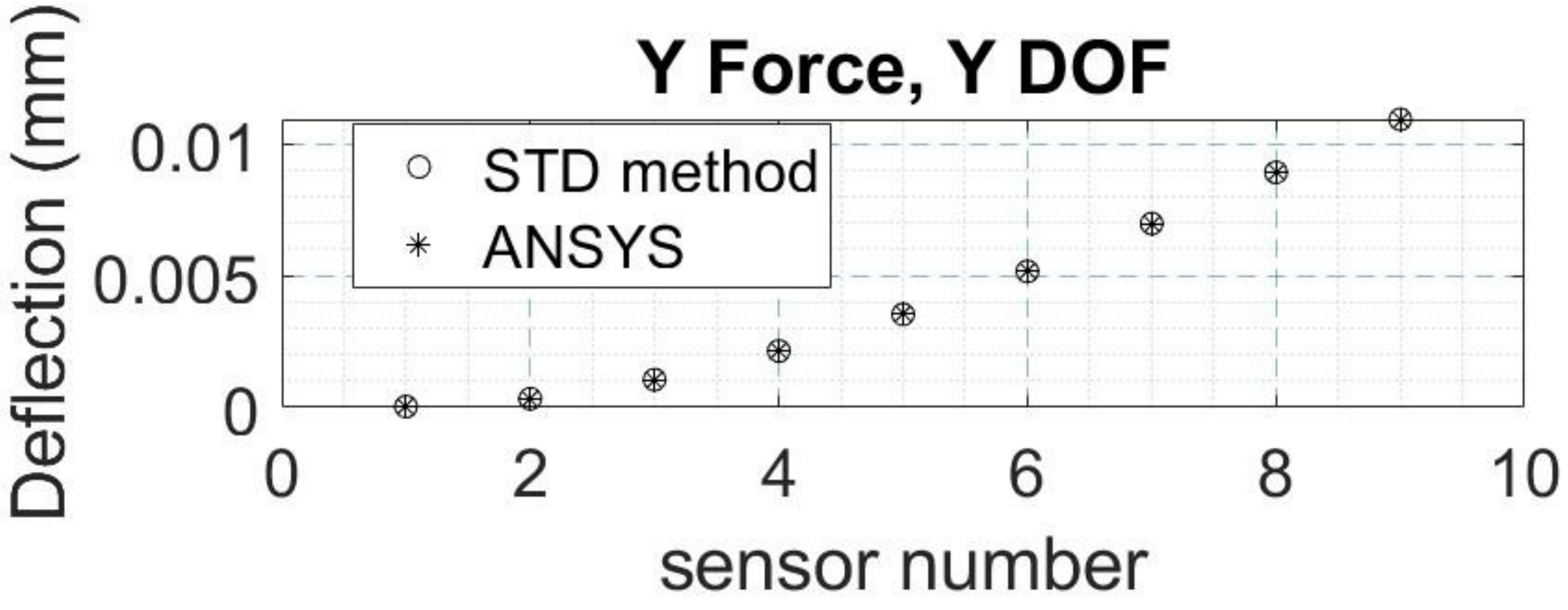
Top Face Sensors, X-Force, and Error (in %)

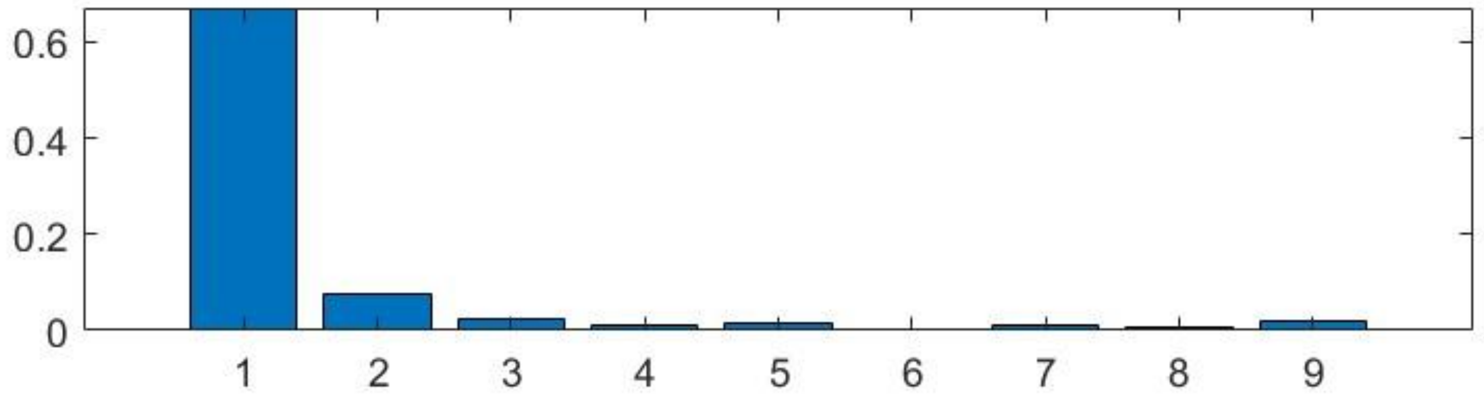


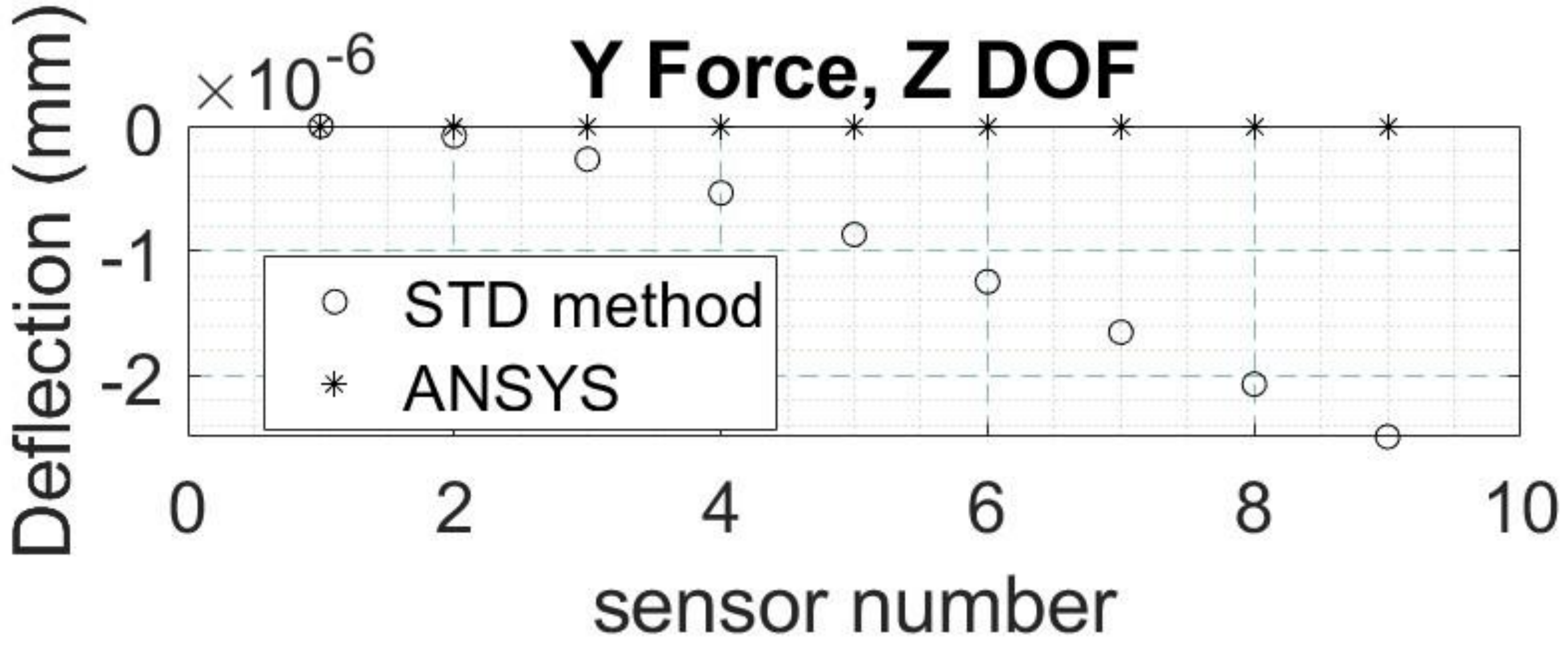
Top Face Sensors, Y-Force, and Error (in %)

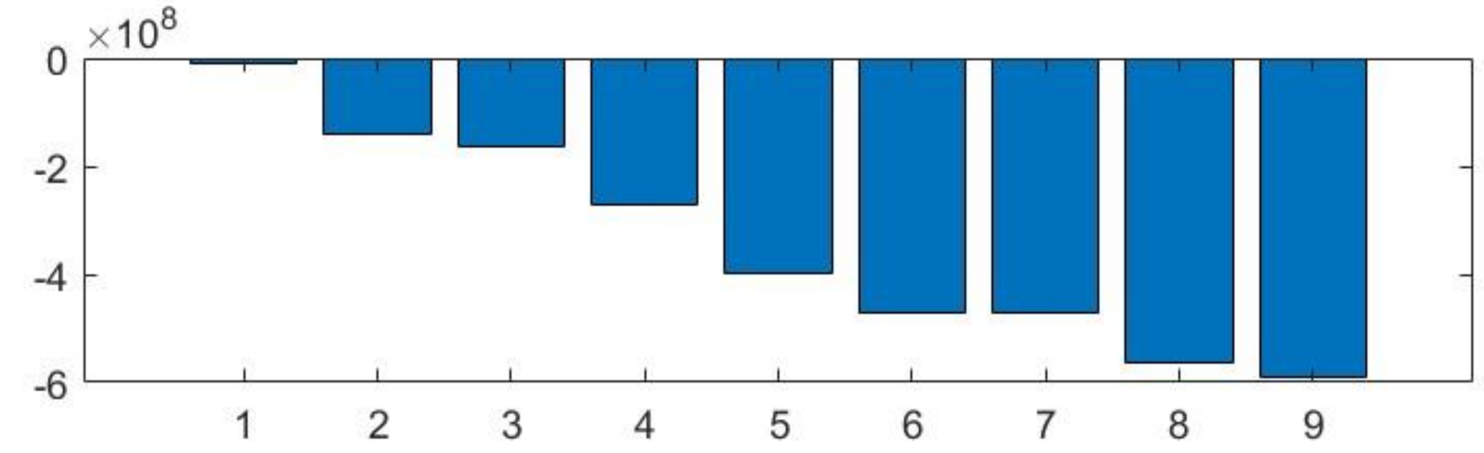












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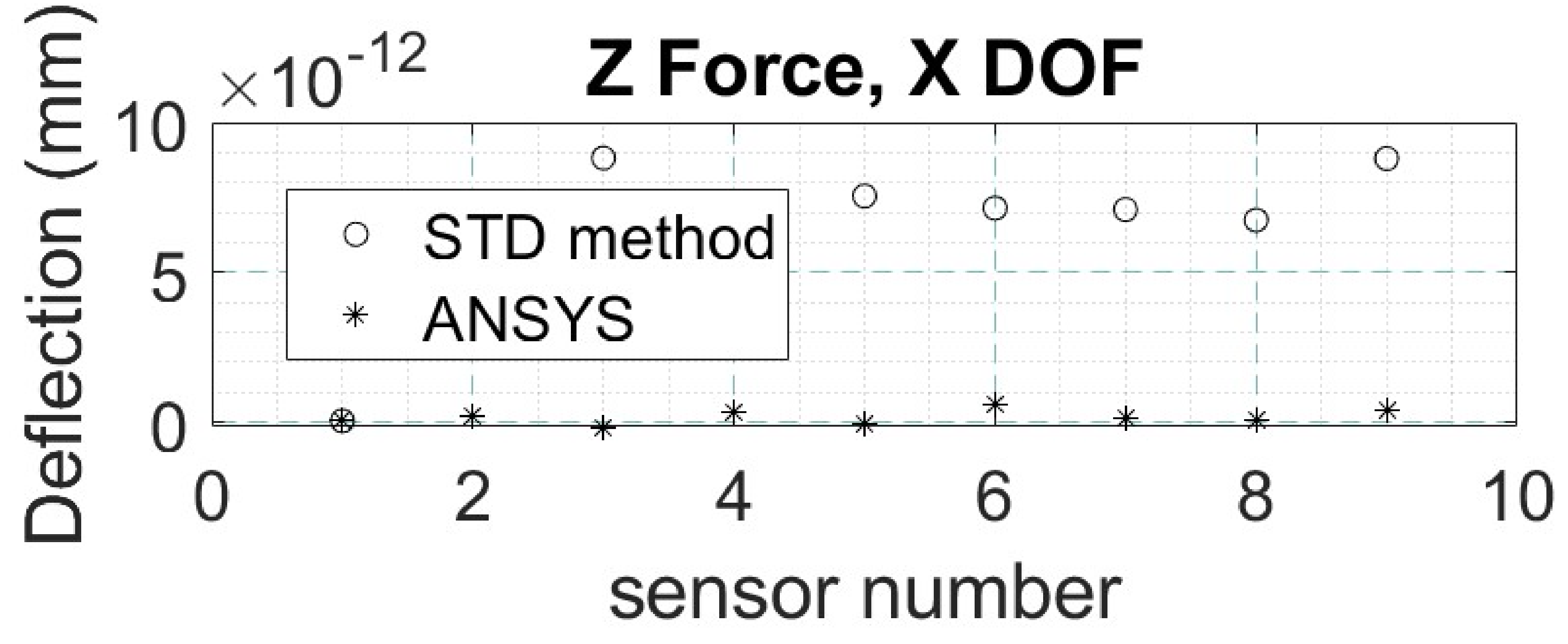
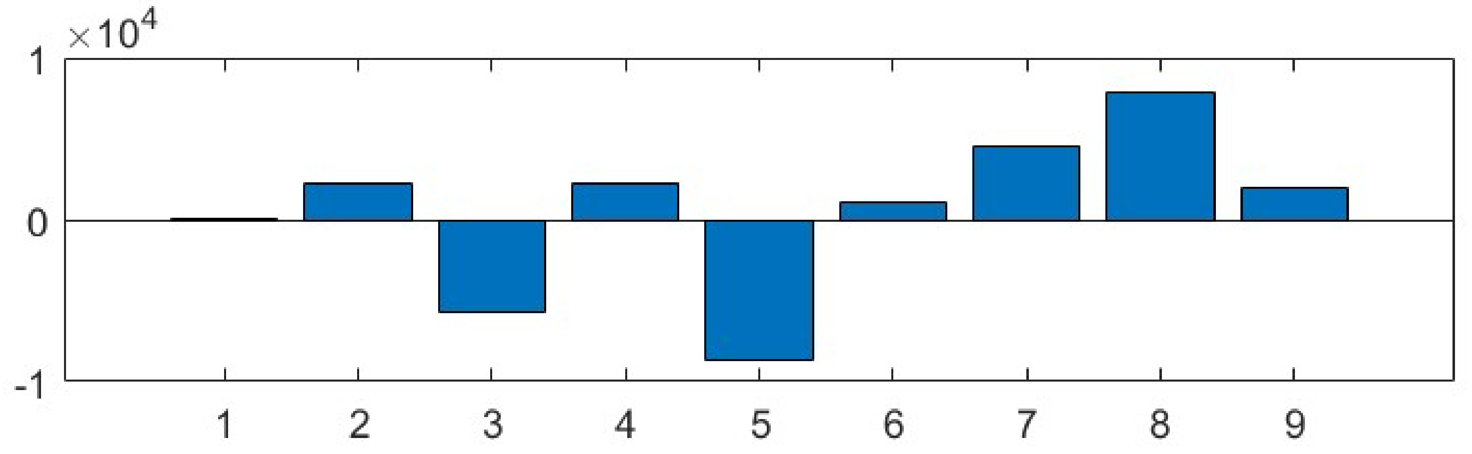
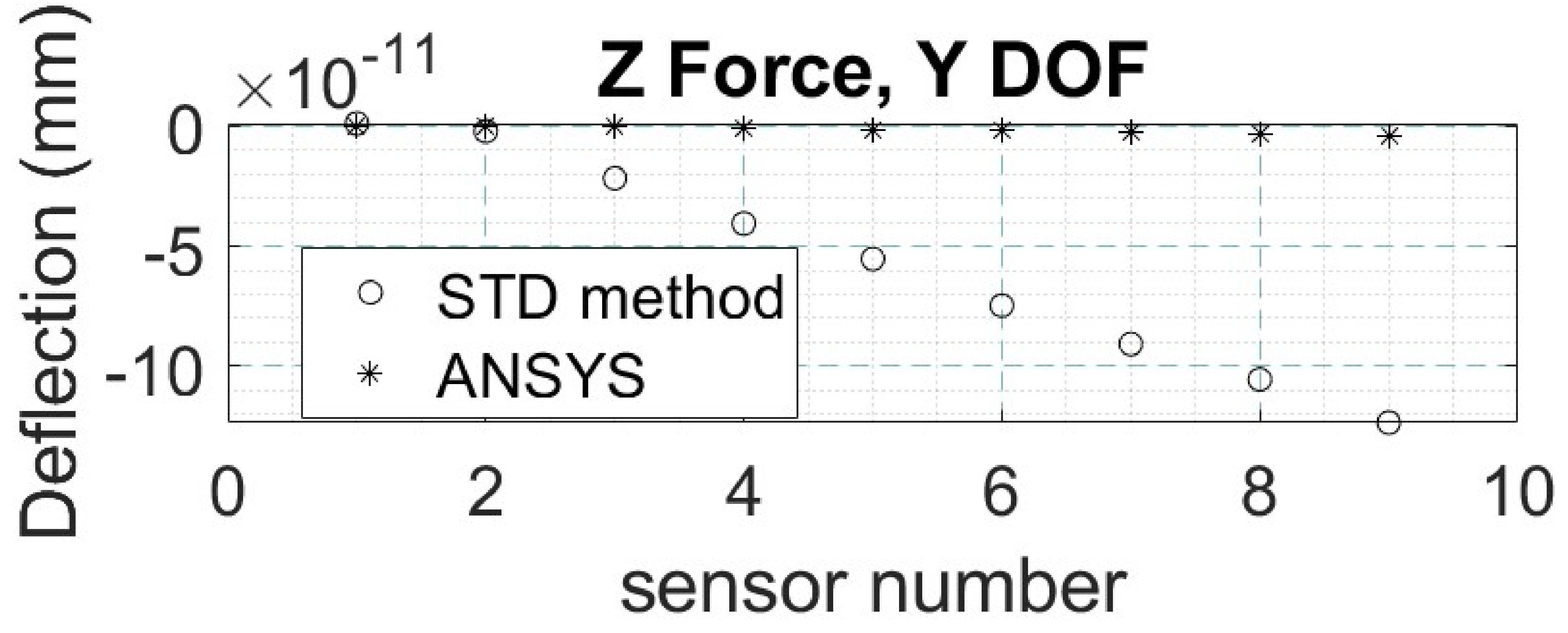
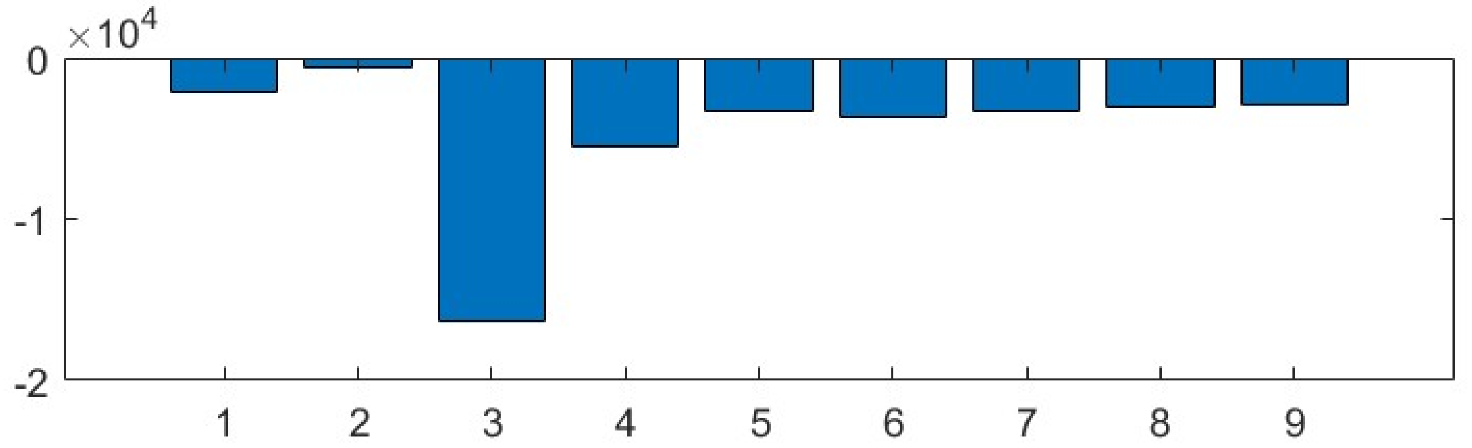
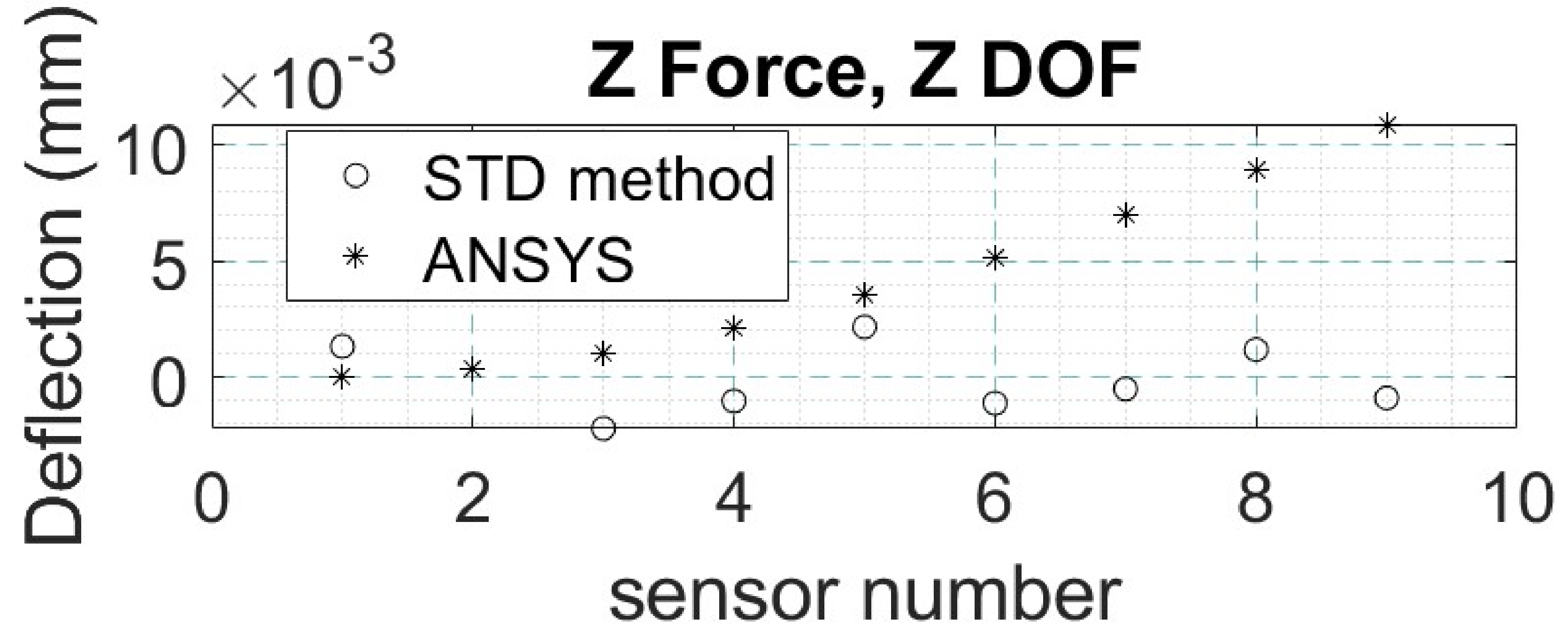
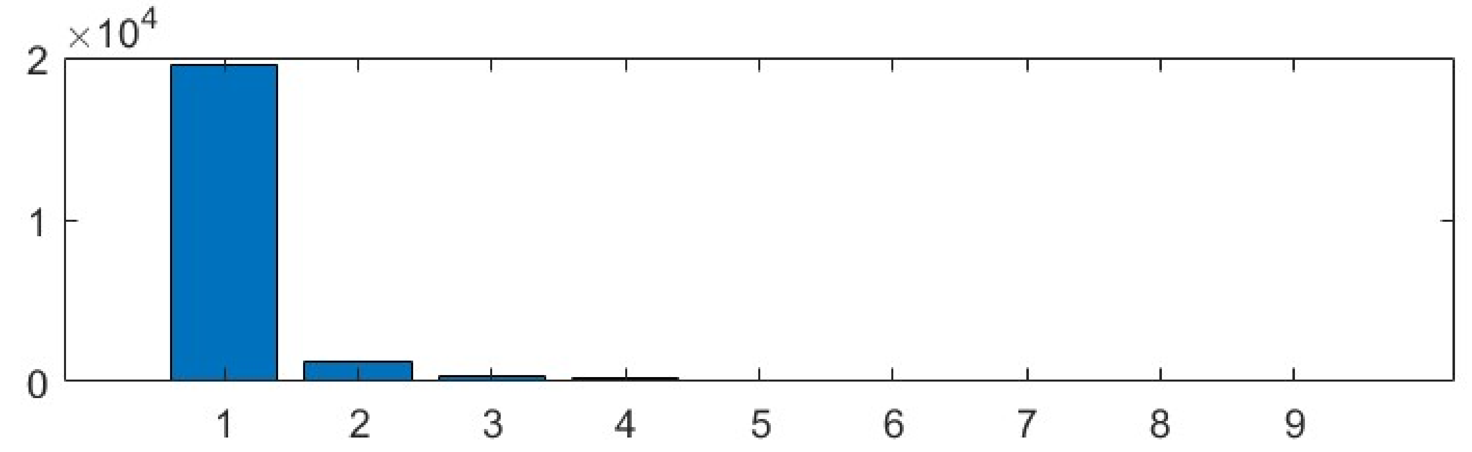




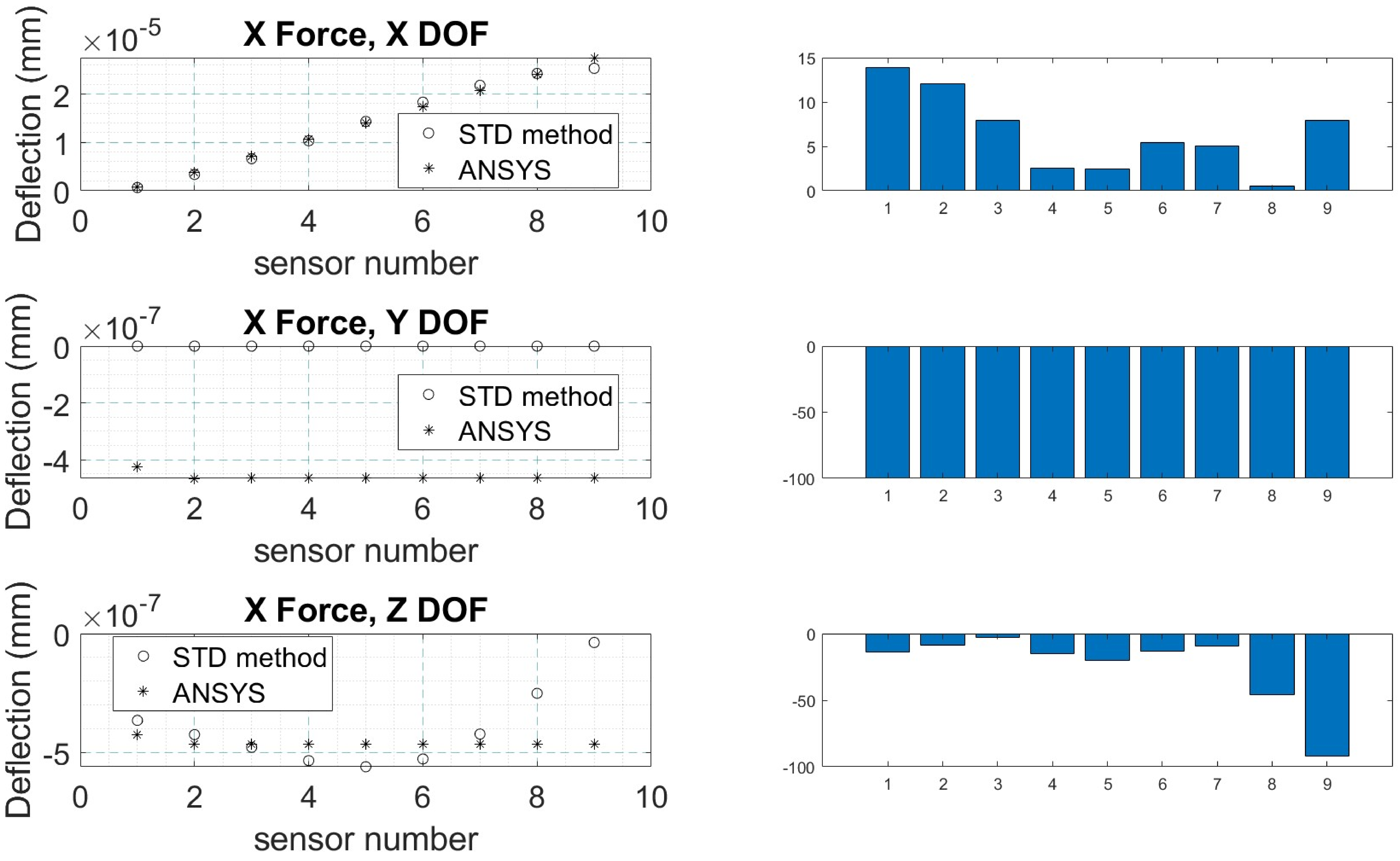








Side Face Sensors, X-Force, and Error (in %)



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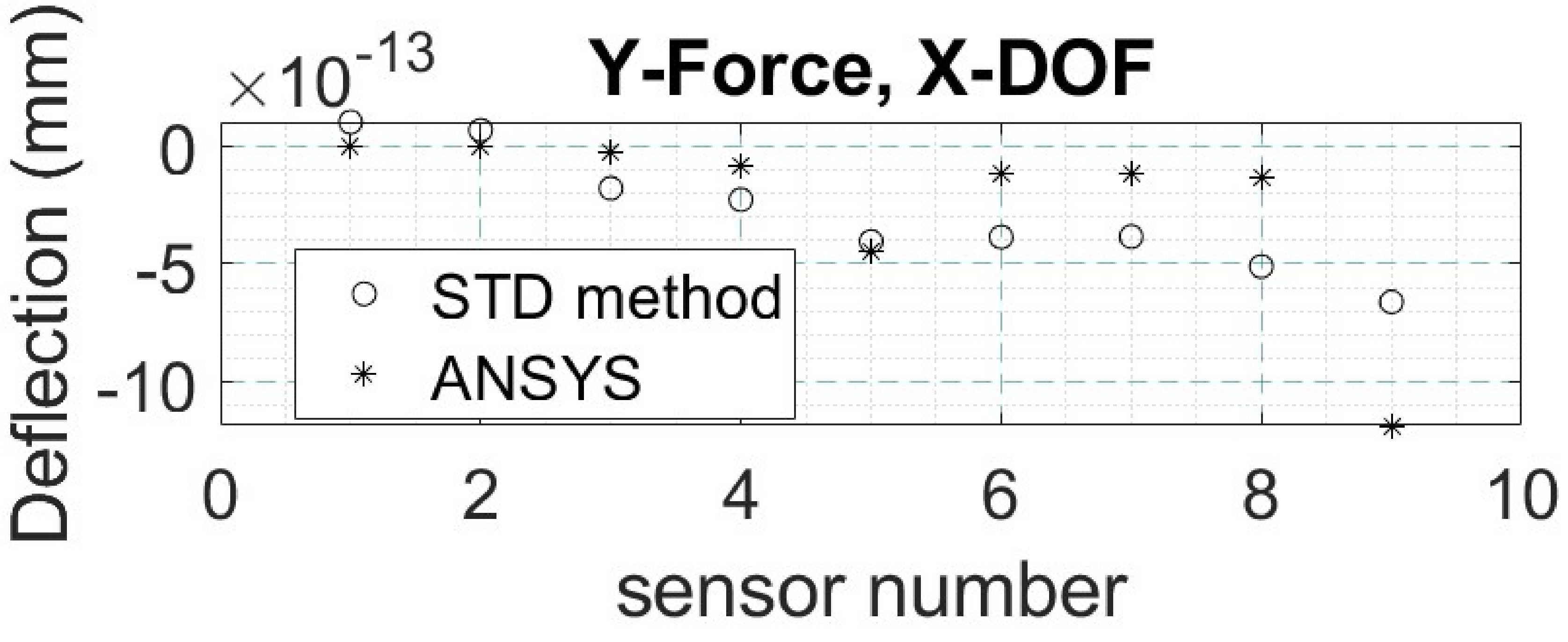
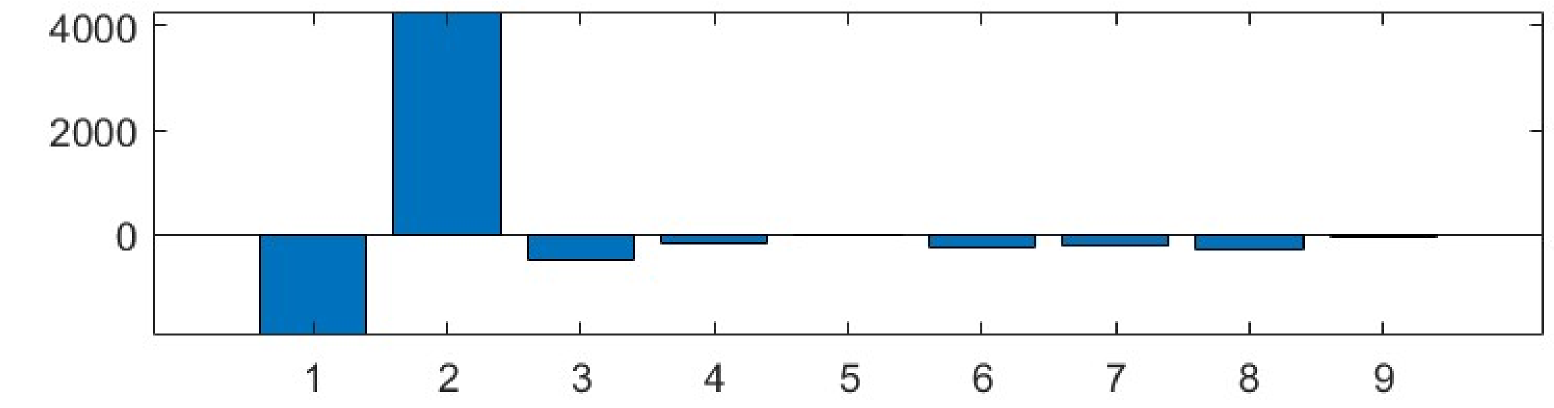
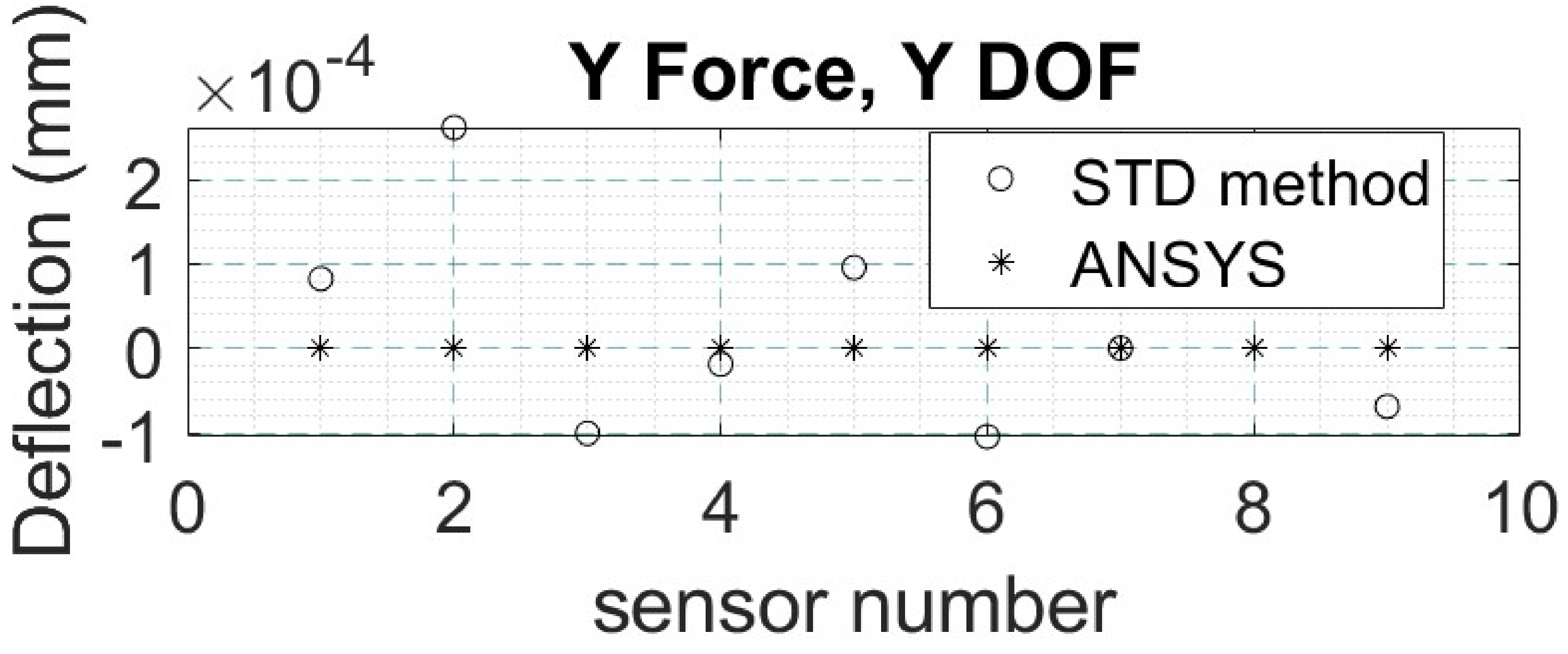
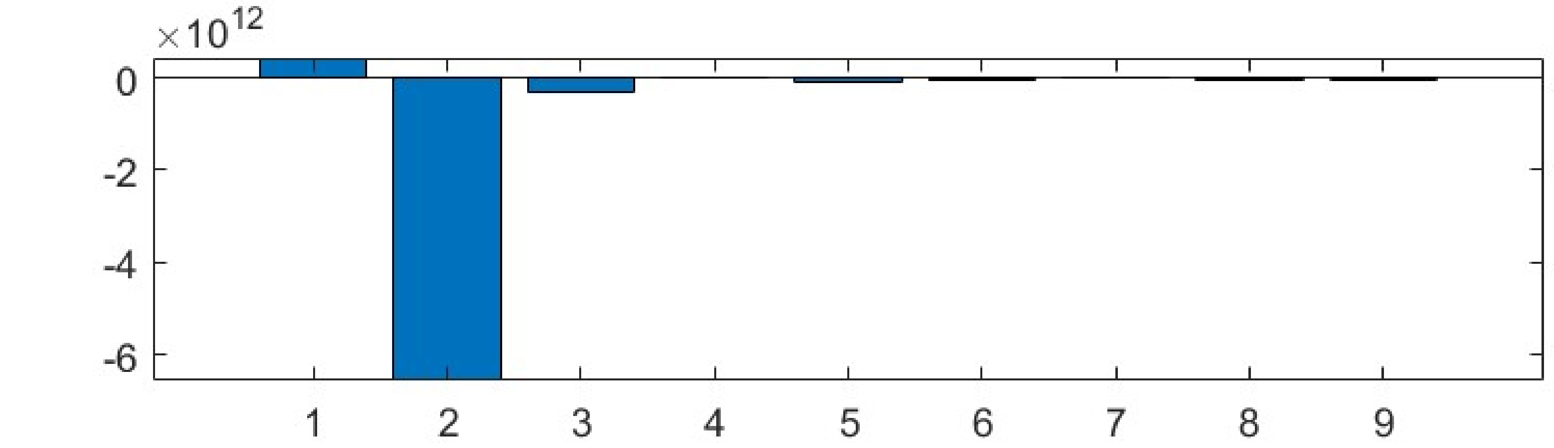
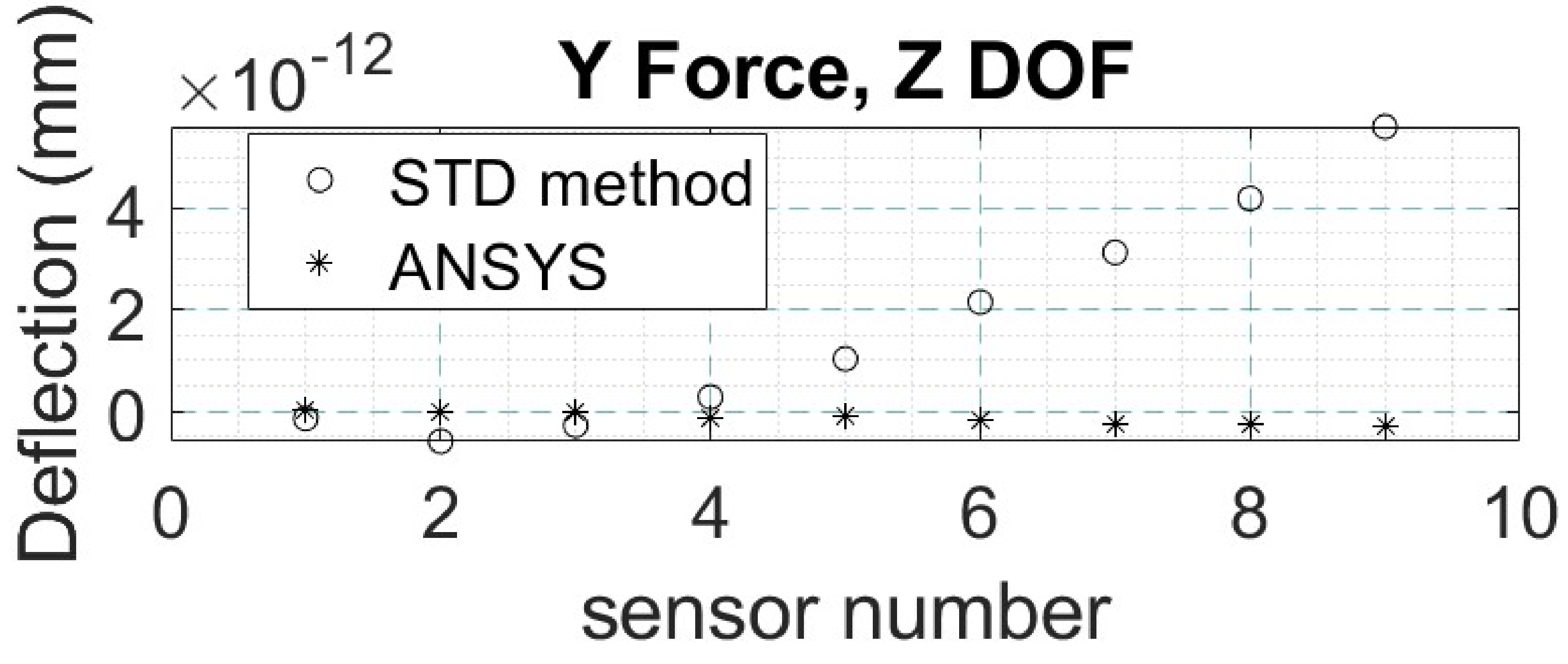
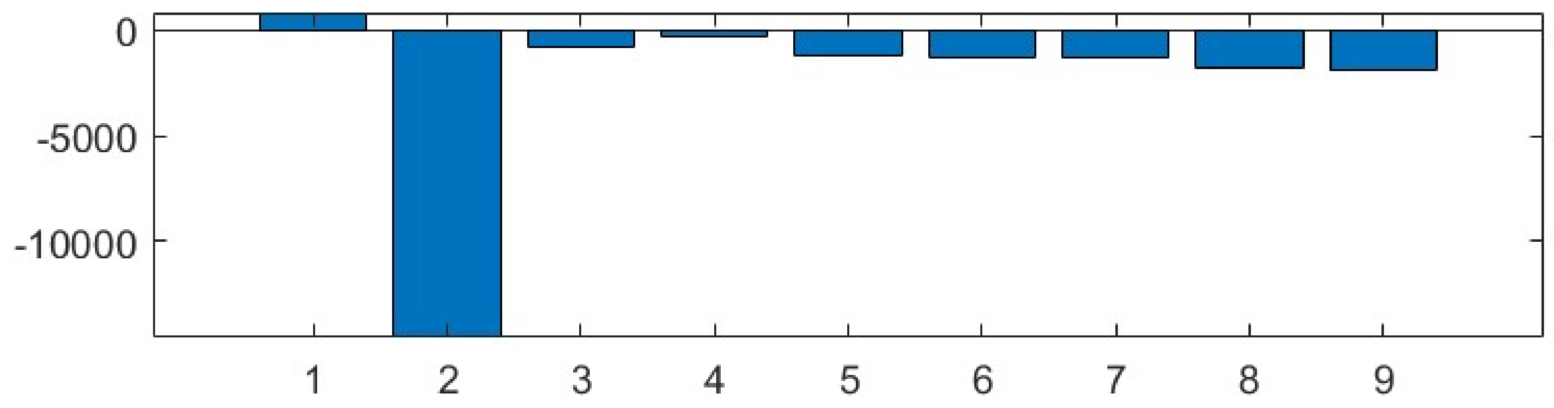




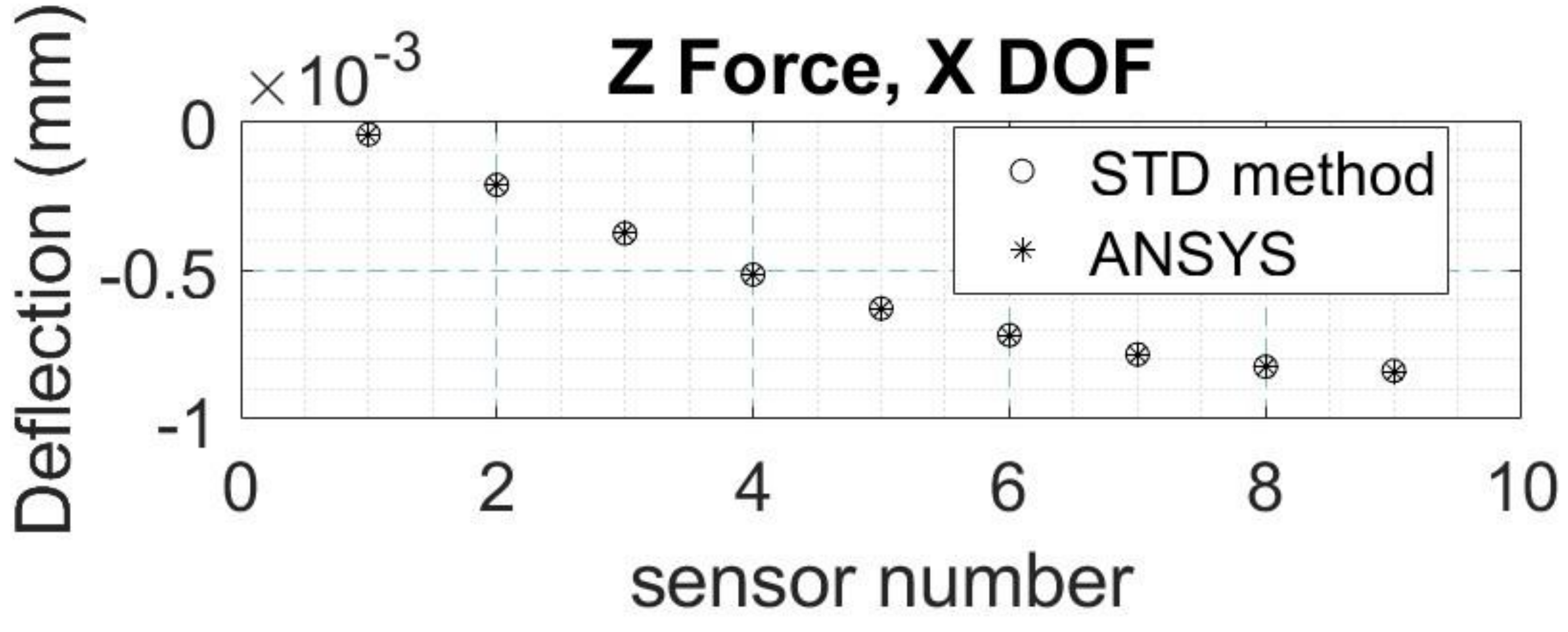


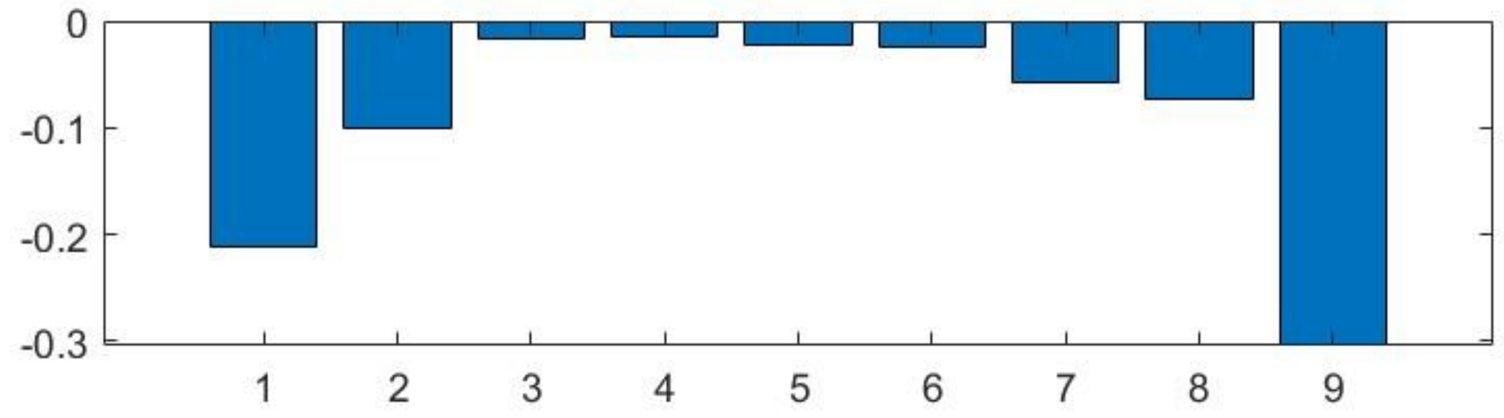


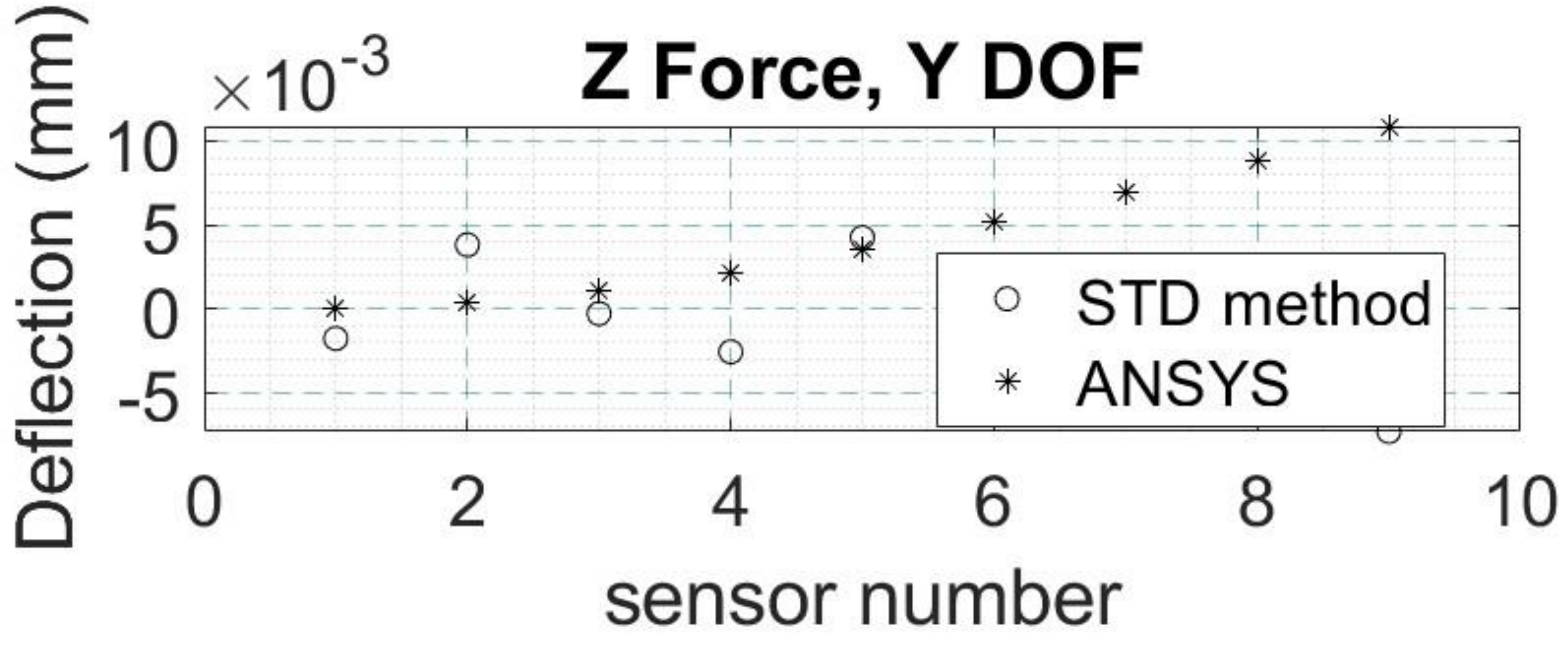


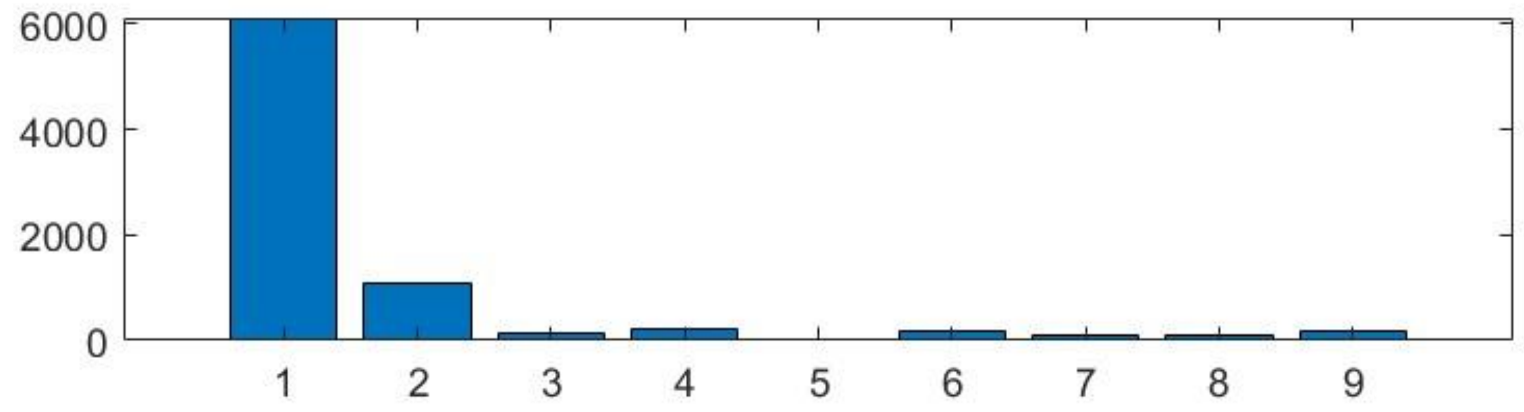


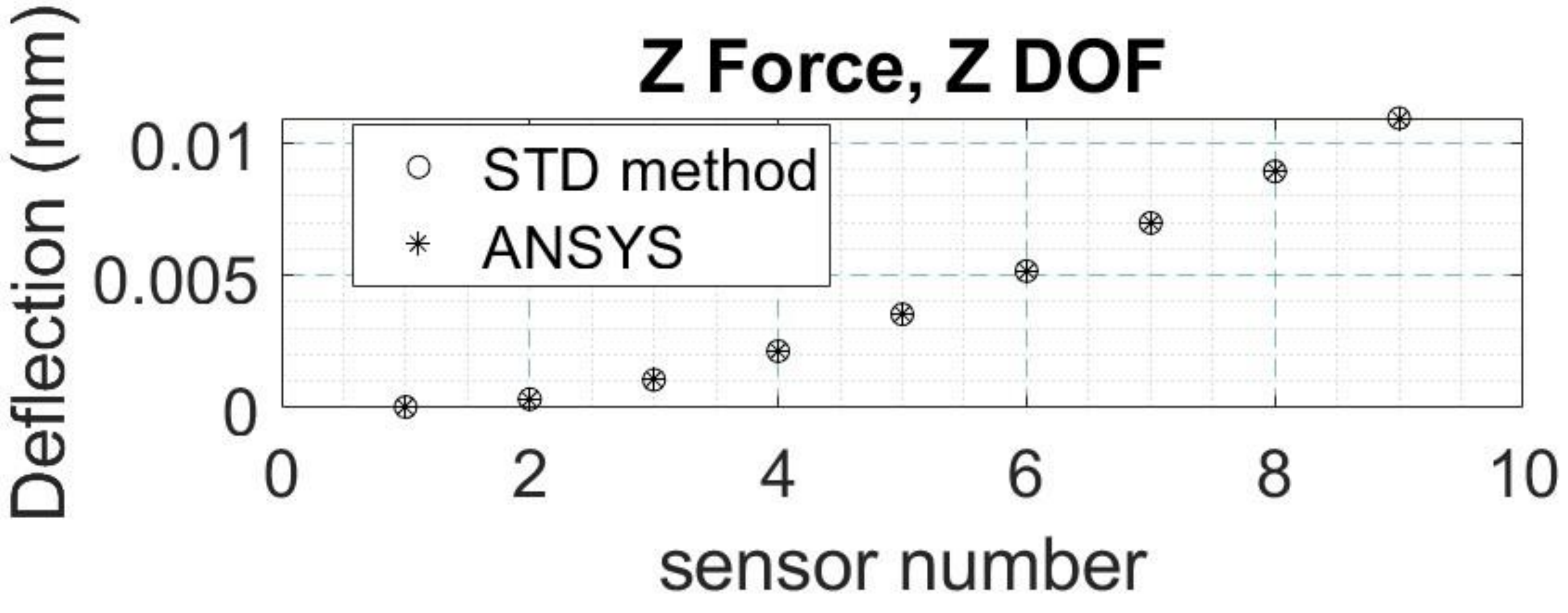
Side Face Sensors, X-Force, and Error (in %)

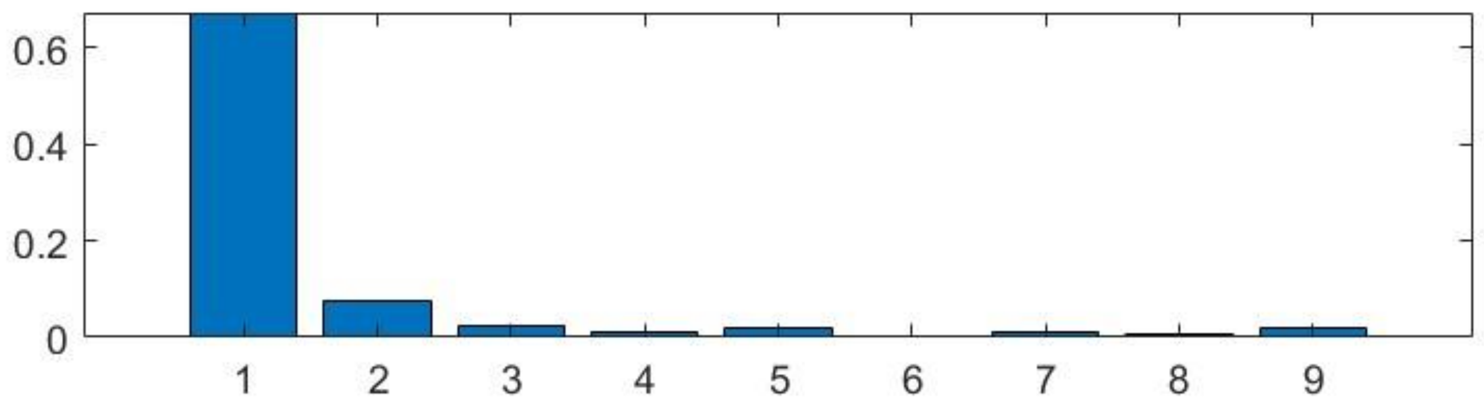












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